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Wazuh SIEM Deployment and Endpoint Monitoring in a Virtual Lab Environment

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1. Introduction

In the previous phase of this project, an Active Directory (AD) environment was successfully set up using a Windows Server as a Domain Controller, along with multiple client systems integrated into the domain. This provided a centralized authentication and identity management system within the lab.

Building upon this environment, the next step is to enhance security visibility by deploying a Security Information and Event Management (SIEM) solution. SIEM systems play a crucial role in modern cybersecurity by providing centralized log collection, monitoring, and real-time threat detection across multiple systems.

Wazuh is an open-source SIEM platform that offers capabilities such as log analysis, intrusion detection, file integrity monitoring, and vulnerability detection. It allows security analysts to monitor endpoints, detect suspicious activities, and respond to potential threats from a centralized dashboard.

In this phase, Wazuh is deployed within the same Active Directory-based lab environment. Agents are installed on domain-joined systems, including Windows Server, Windows 10 clients, and Ubuntu client, enabling centralized monitoring of system activity and security events.

This implementation demonstrates how SIEM tools integrate with enterprise environments and provides hands-on experience with real-world security monitoring.

The objectives of this phase are:

- Deploy Wazuh SIEM in an existing Active Directory lab environment
- Configure and access the Wazuh dashboard
- Install and configure agents on domain-joined systems
- Monitor logs and security events from multiple endpoints
- Understand how SIEM tools detect and analyze potential threats

2. Wazuh Overview & Architecture

2.1 What is Wazuh?

Wazuh is an open-source security platform used for threat detection, security monitoring, and incident response. It functions as a Security Information and Event Management (SIEM) solution by collecting and analyzing logs from multiple systems in real time.

In this project, Wazuh is deployed within an Active Directory-based lab environment to monitor domain activity, detect suspicious behavior, and simulate real-world enterprise security operations.

Wazuh provides capabilities such as:

- Log collection and analysis
- Intrusion detection
- File integrity monitoring
- Vulnerability detection
- Security alerting and reporting

By integrating Wazuh into the lab, we create a centralized system that enhances visibility across all endpoints.

2.2 Core Components of Wazuh

Wazuh consists of several key components that work together to provide end-to-end security monitoring.

1. **Wazuh Agents:** Wazuh agents are lightweight software installed on endpoints such as Windows and Linux systems. Their primary role is to:
    - Collect system logs and security events
    - Monitor file changes and system activity

- Send collected data to the Wazuh Manager

In this lab, agents are installed on:

- Windows Server (Domain Controller)
- Windows 10 Client
- Ubuntu Client

2. **Wazuh Manager:** The Wazuh Manager is the central component responsible for processing data received from agents. Its functions include:
   - Receiving logs from multiple endpoints
   - Analyzing data using detection rules
   - Identifying suspicious or malicious activity
   - Generating security alerts

   The manager acts as the brain of the SIEM system, performing all analysis and decision-making.

3. **Wazuh Indexer:** The Wazuh Indexer is responsible for storing and indexing processed data. Its key roles are:
   - Storing logs and alerts
   - Enabling fast search and retrieval
   - Supporting historical analysis

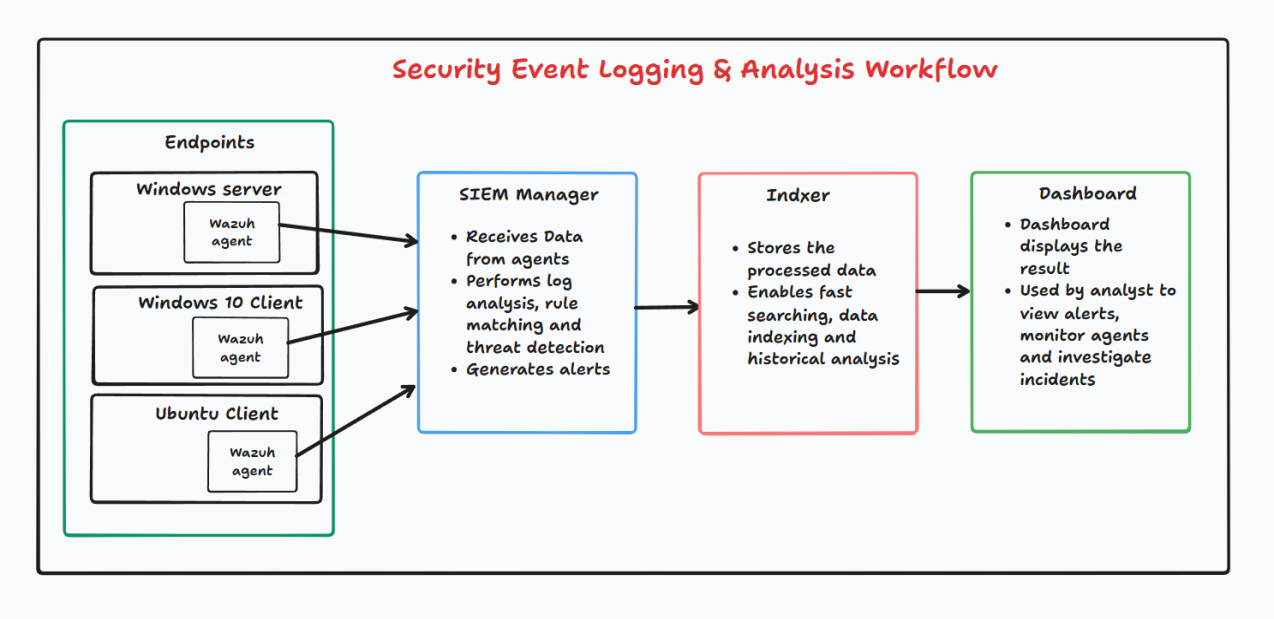
   This component ensures that large volumes of data can be efficiently managed and queried when needed.

4. **Wazuh Dashboard:** The Wazuh Dashboard provides a graphical interface for interacting with the system. It allows users to:
   - View security alerts
   - Monitor agent status
   - Analyze logs
   - Investigate incidents

The dashboard is the primary interface used by security analysts to monitor and respond to threats.

2.3 How Wazuh Works (Data Flow)

The Wazuh platform operates through a structured workflow that ensures efficient collection, processing, storage, and visualization of security events.



Workflow Explanation:

1. **Event Generation at Endpoints:** Endpoints generate logs related to system activity, authentication, and security events. Wazuh agents continuously monitor and collect this data.
2. **Data Transmission to Wazuh Manager:** The collected logs are securely forwarded to the Wazuh Manager for processing.
3. **Log Analysis and Threat Detection:** The manager analyzes incoming data using predefined rules and detection logic to identify suspicious or malicious behavior.
4. **Data Storage in Indexer:** Processed logs and alerts are stored in the Wazuh Indexer, enabling fast search and historical analysis.

5. **Visualization in Dashboard:** The Wazuh Dashboard displays alerts and monitoring data, allowing users to investigate and respond to incidents.

3. Lab Architecture

3.1 Lab Environment Overview

The Wazuh deployment is built on top of the previously configured Active Directory lab environment. The setup is designed to simulate a real-world enterprise network where multiple endpoints are centrally monitored using a SIEM solution.

The lab consists of the following systems:

- Windows Server (Domain Controller): Hosts Active Directory services and manages domain authentication.
- Windows 10 Client: Joined to the domain and used to simulate user activity.
- Ubuntu Client: Integrated with Active Directory and used for cross-platform monitoring.
- Wazuh Server (Ubuntu-based): Hosts Wazuh Manager, Indexer and Dashboard.

All systems are connected within the same internal network created using VirtualBox.

3.2 Network Configuration

The lab uses a NAT Network in VirtualBox to ensure isolated communication between all virtual machines.

- Network Name: Ritik_SOC
- IP Range: 172.16.1.0/24
- DHCP: Enabled

Each machine is assigned an IP address within this range, enabling seamless communication between:

- Domain Controller (Windows Server)

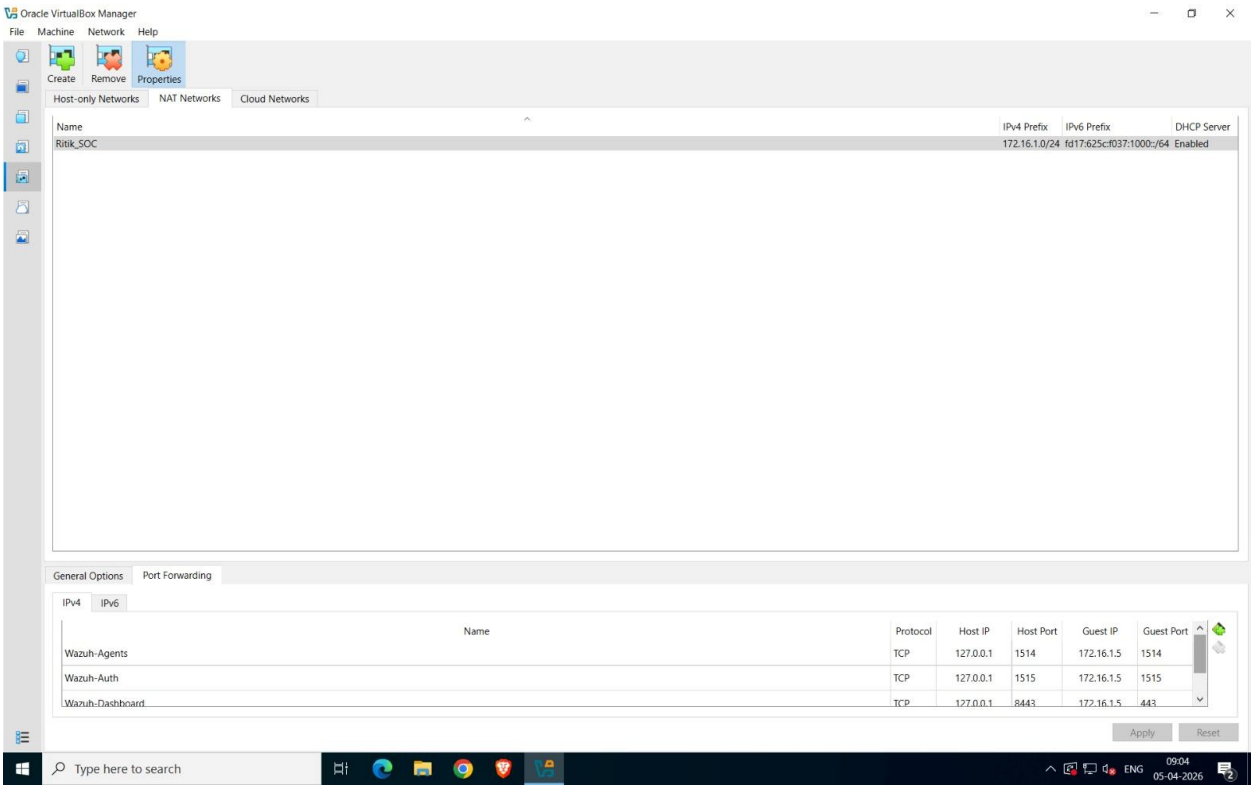
- Clients
- Wazuh Server

This setup ensures that all logs and events can be transmitted securely within the lab environment.

3.3 Port Forwarding Configuration

The Wazuh dashboard can be accessed from all the clients and servers are in virtual box but all the systems are slow as they have limited memory, CPU's and storage. So main or host machine is used to access the Wazuh Dashboard.

To access the Wazuh Dashboard from the host machine (main system browser), port forwarding rules were configured in VirtualBox.



The following port forwarding rules were created:

Service	Host Port	Guest IP	Guest Port
Wazuh Agents	1514	172.16.1.5	1514
Wazuh Auth	1515	172.16.1.5	1515
Wazuh Dashboard	8443	172.16.1.5	443

- Agent Port (1514): Used by Wazuh agents to send logs to the Wazuh Manager.
- Authentication Port (1515): Used for agent registration and authentication.
- Dashboard Access (8443): Allows access to the Wazuh Dashboard from the host machine using https://127.0.0.1:8443

Port forwarding enables interaction with the Wazuh web interface without requiring direct access to the VM's internal IP.

4. Wazuh Deployment

Wazuh Server Installation (OVA Setup)

The Wazuh server was deployed using the official Wazuh OVA (Open Virtual Appliance) file. This method simplifies installation by providing a pre-configured environment with all required components.

Step 1: Import Wazuh OVA into VirtualBox

The Wazuh Virtual Machine was created by importing the OVA file:

- File used: wazuh-4.14.4.ova (sha512)
- wazuh-4.14.4.ova (sha512)

Steps:

1. Double Click the file
2. Click on settings and select your Machine Based Folder. Also set MAC Address Policy to “Include Only NAT-Network Adapter…”
3. Click Finish

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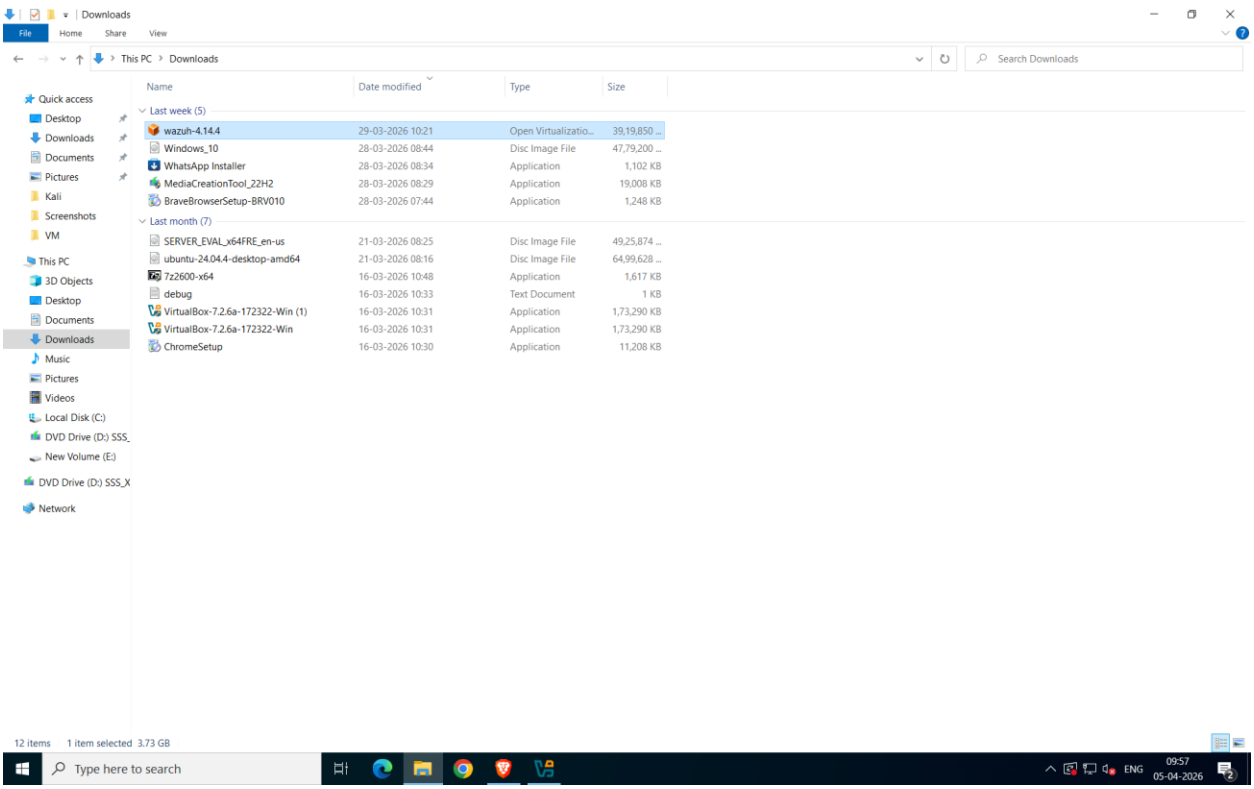

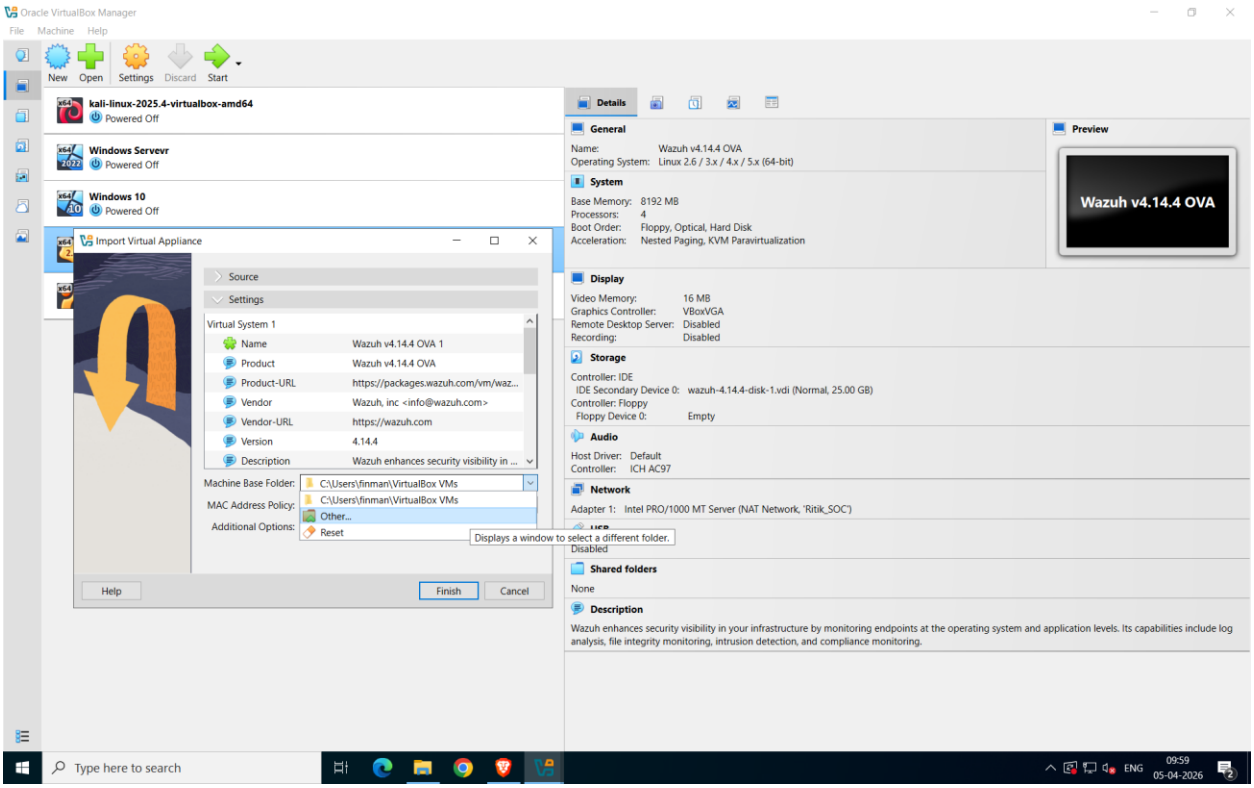

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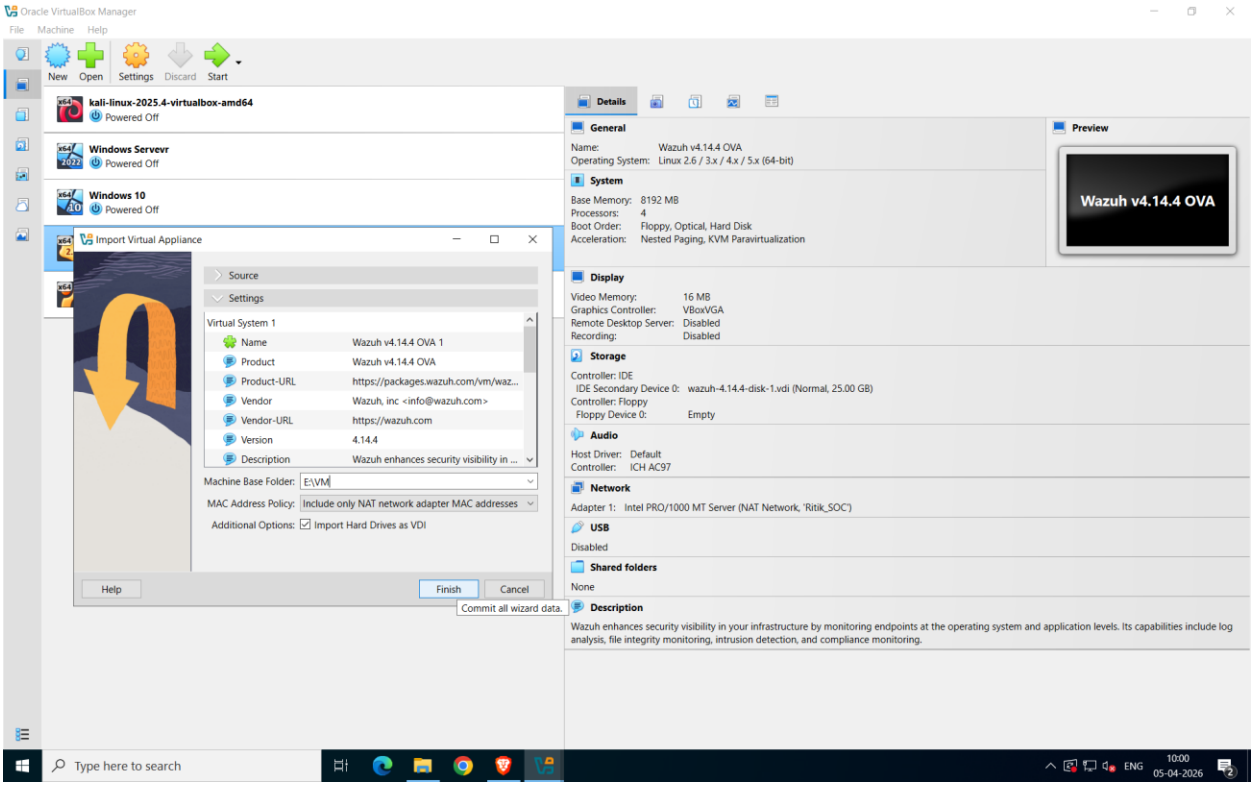

Step 2: Wazuh Server Setup

After importing:

- The Wazuh VM was started
- The system booted with pre-installed services Wazuh Manager, Wazuh Indexer and Wazuh Dashboard
- Login to wazuh using default username: wazuh-user and password: wazuh

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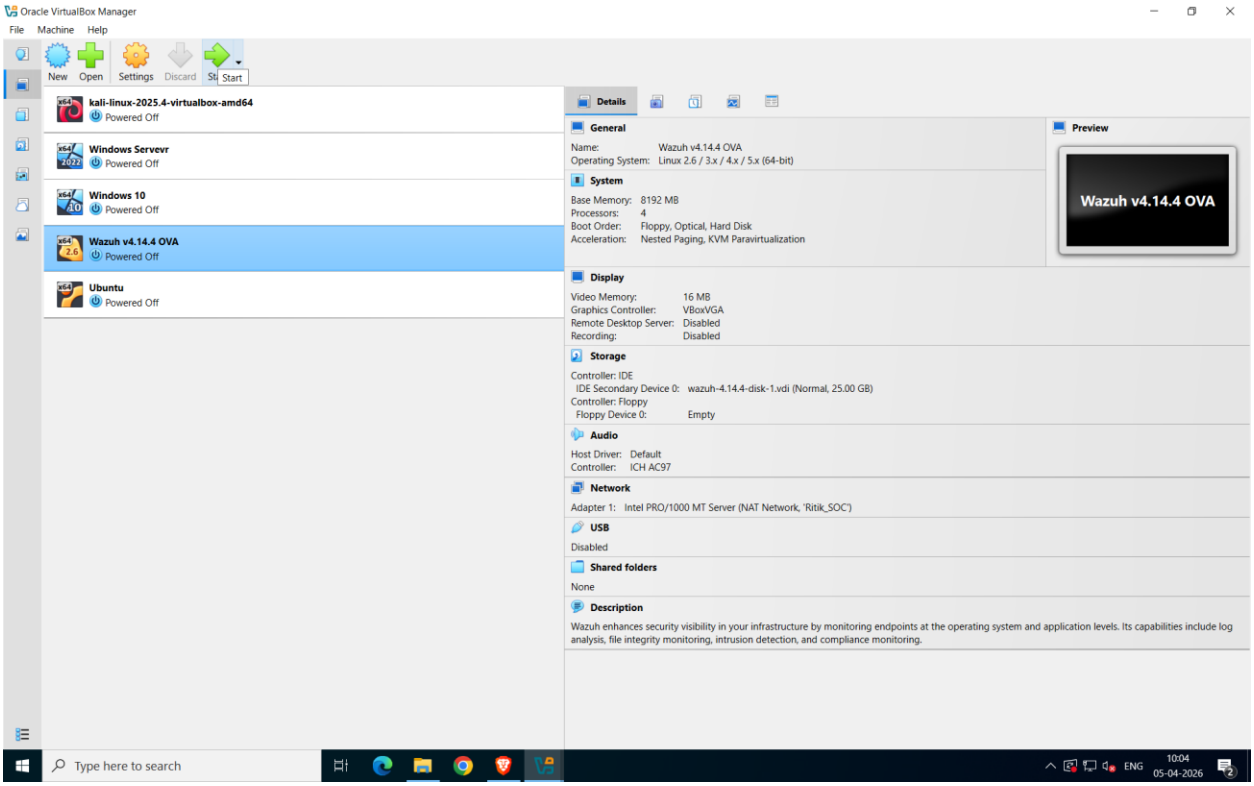

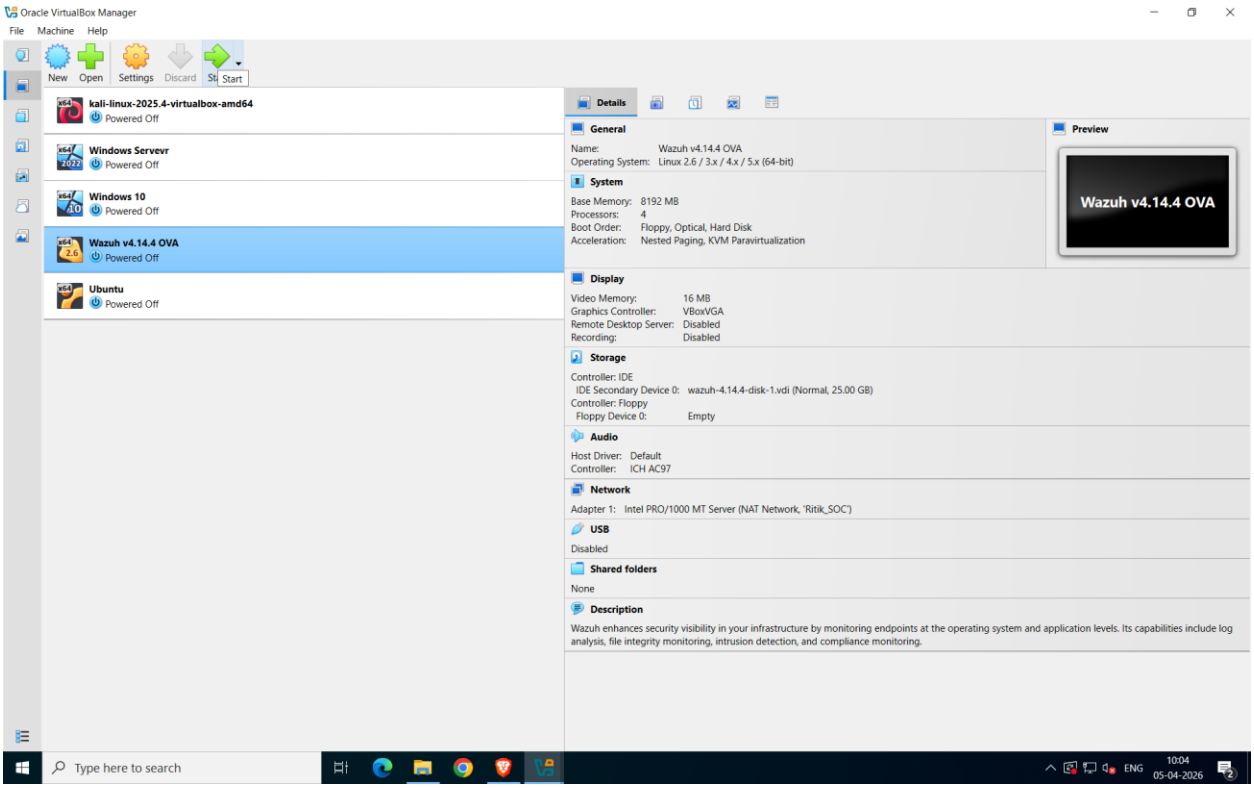

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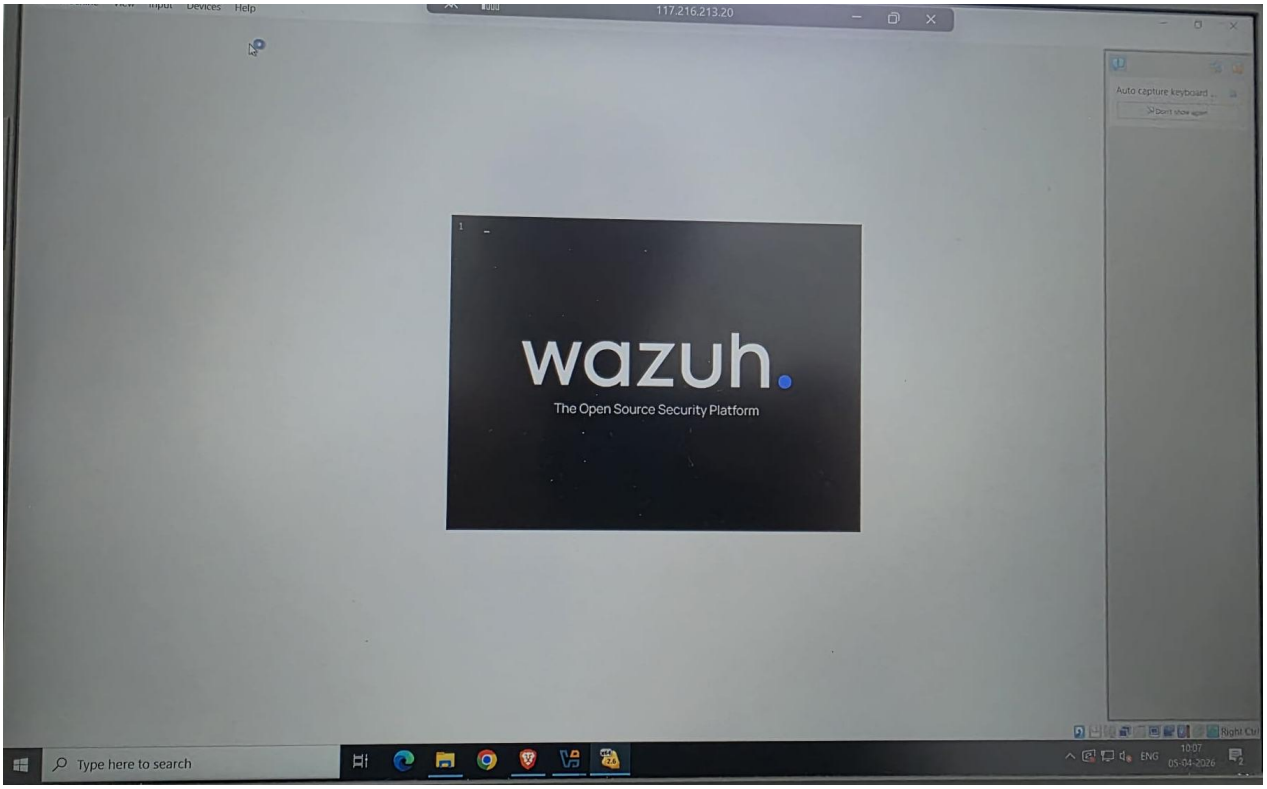

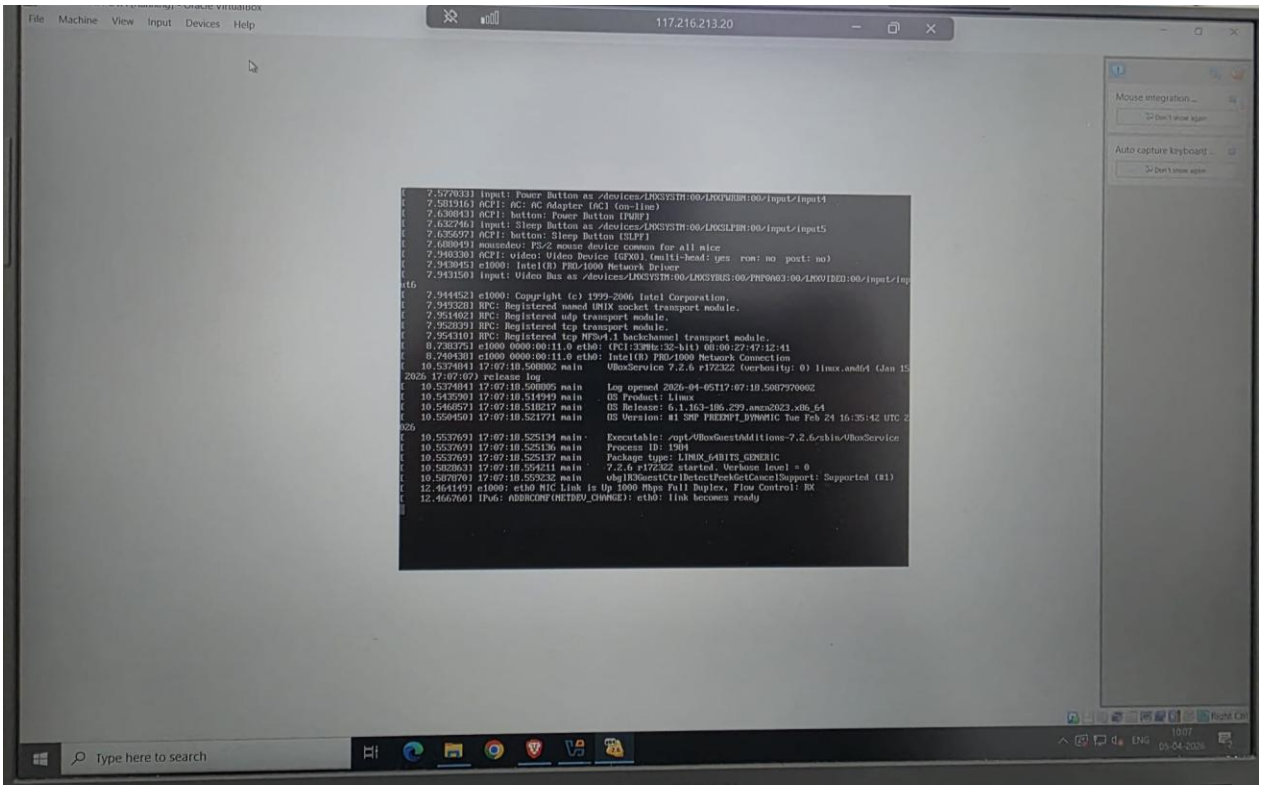

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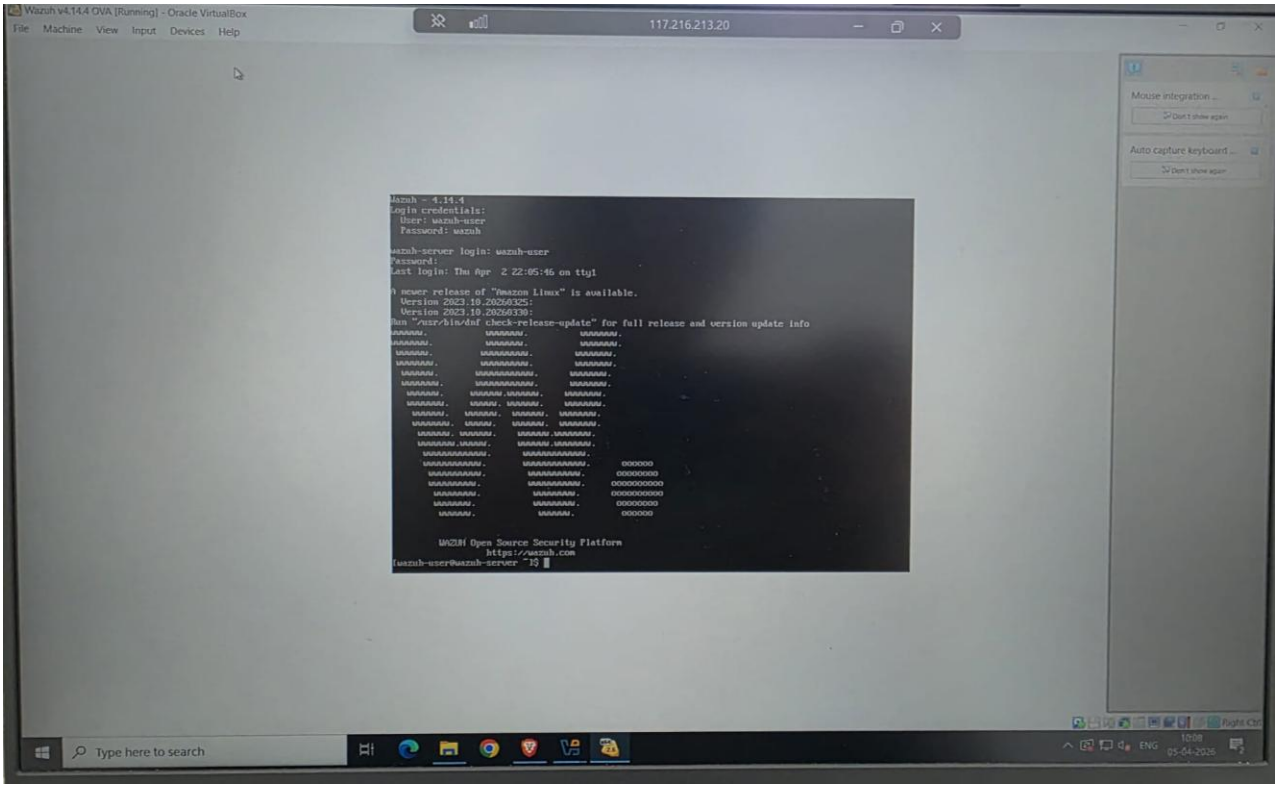
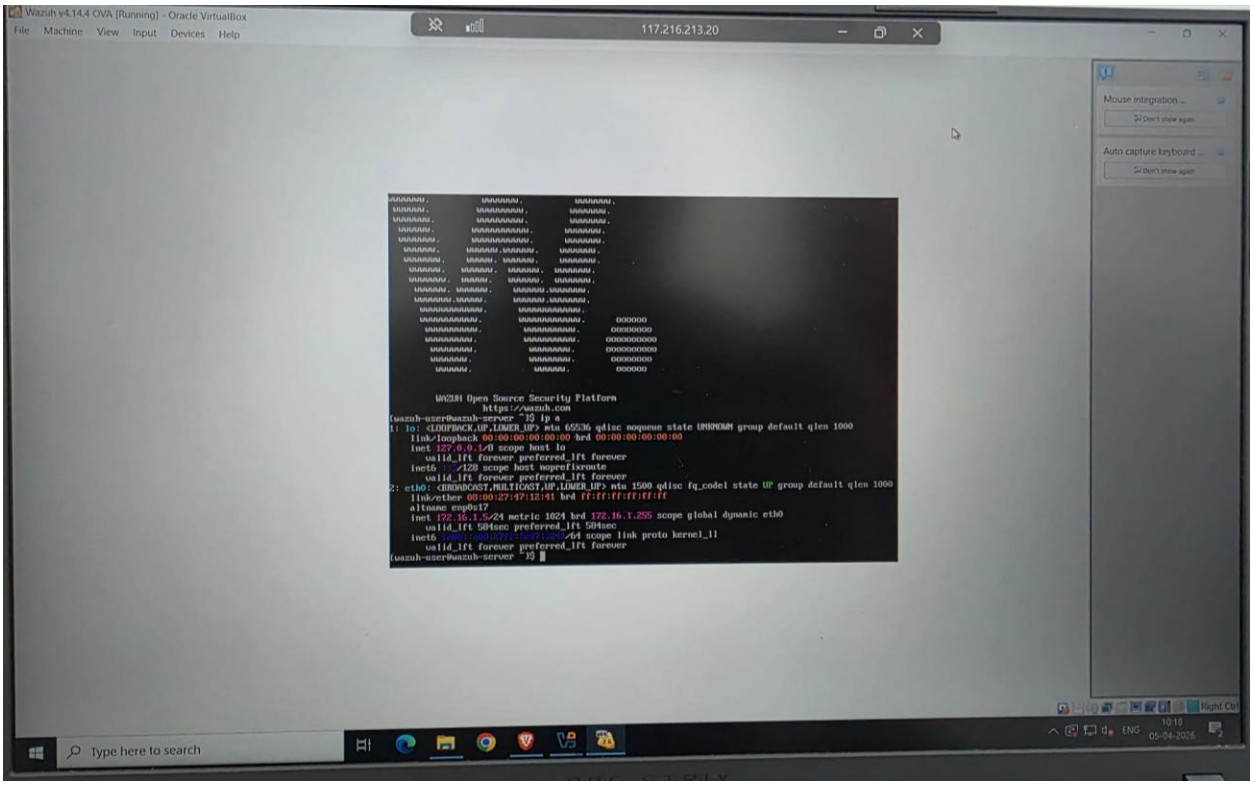


Step 3: Network Configuration

The Wazuh server was connected to the same NAT Network (Ritik_SOC) to communicate with other lab machines.

- Assigned IP address: 172.16.1.5

This allows agents to connect to the manager and also allows communication between AD and clients.



Step 4: Port forwarding Configuration

To access the Wazuh Dashboard from the host machine (main system browser), port forwarding rules were configured in VirtualBox.

The following port forwarding rules were created:

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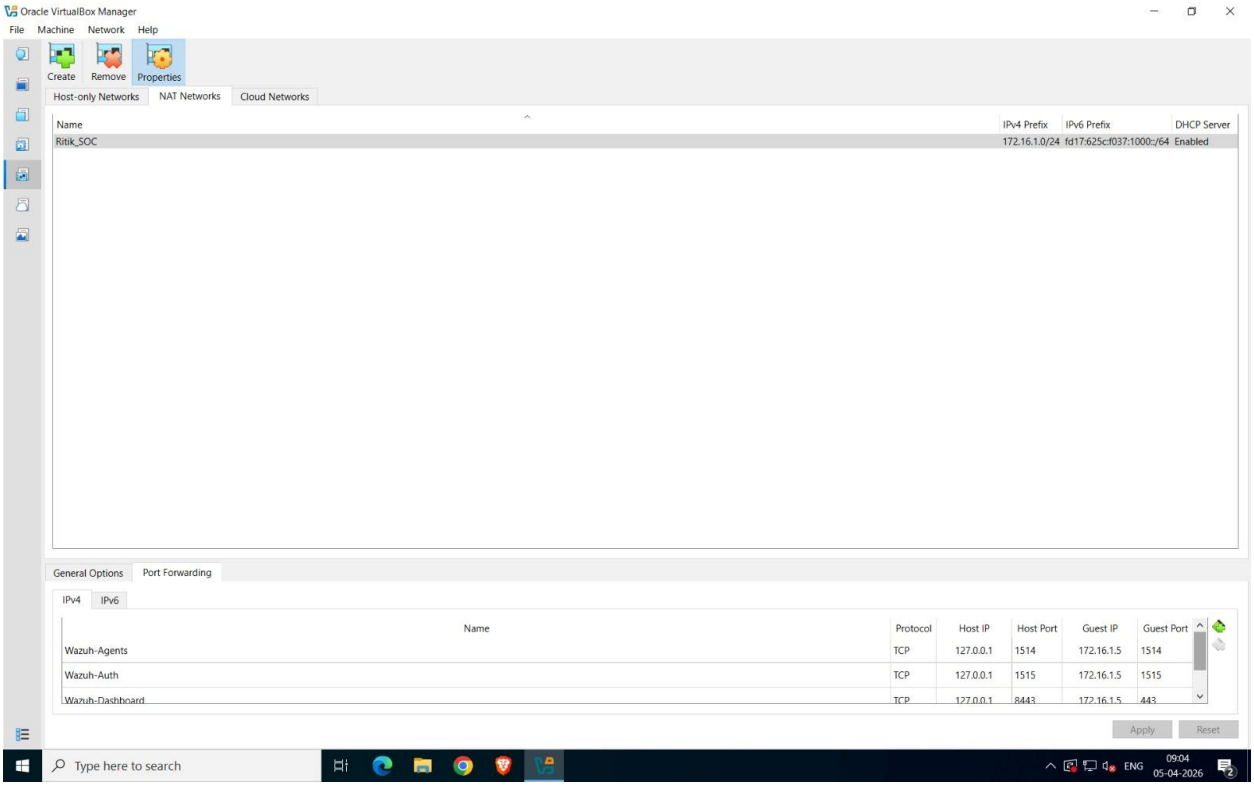

This port forwarding rules allows us to access Wazuh Dashboard from the host machine browser.

Step 5: Access Wazuh Dashboard

- Open host or main machine browser
- Paste the following url: https://127.0.0.1:8443/app/login
- Login using default username: admin and password: admin

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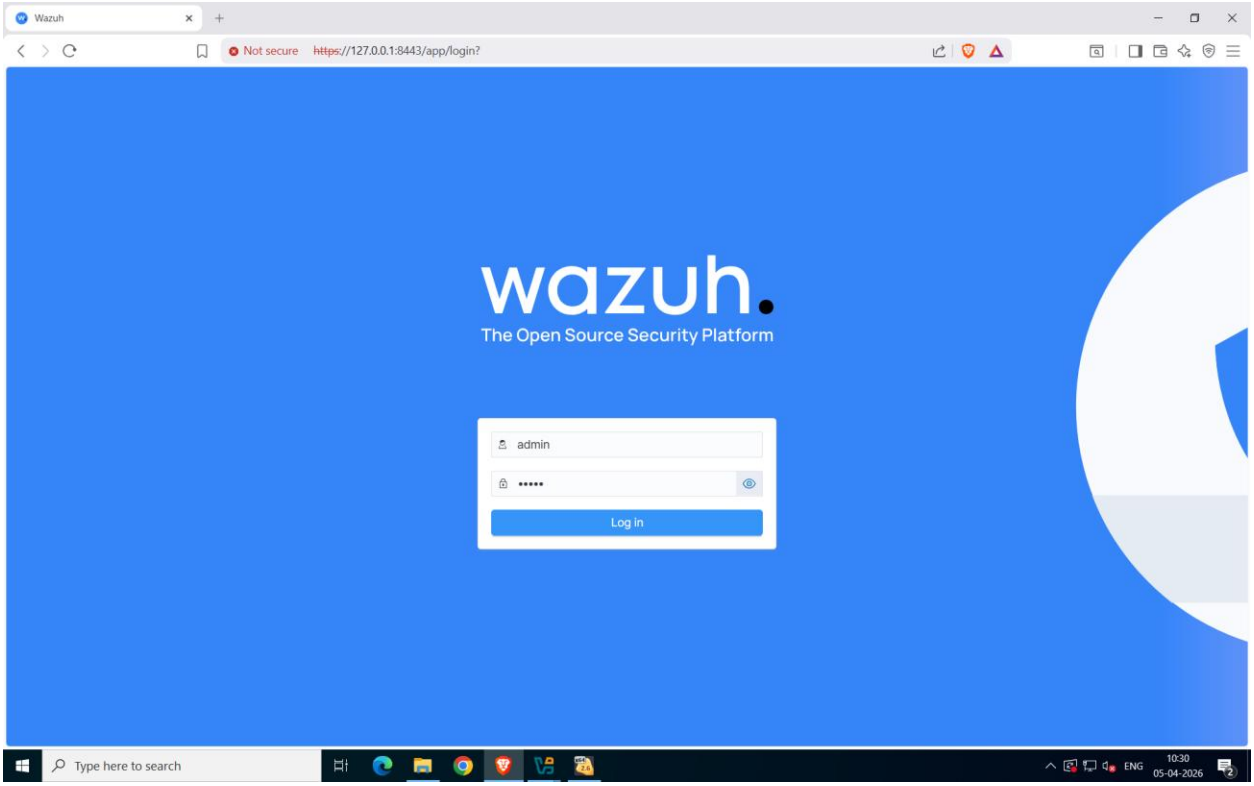

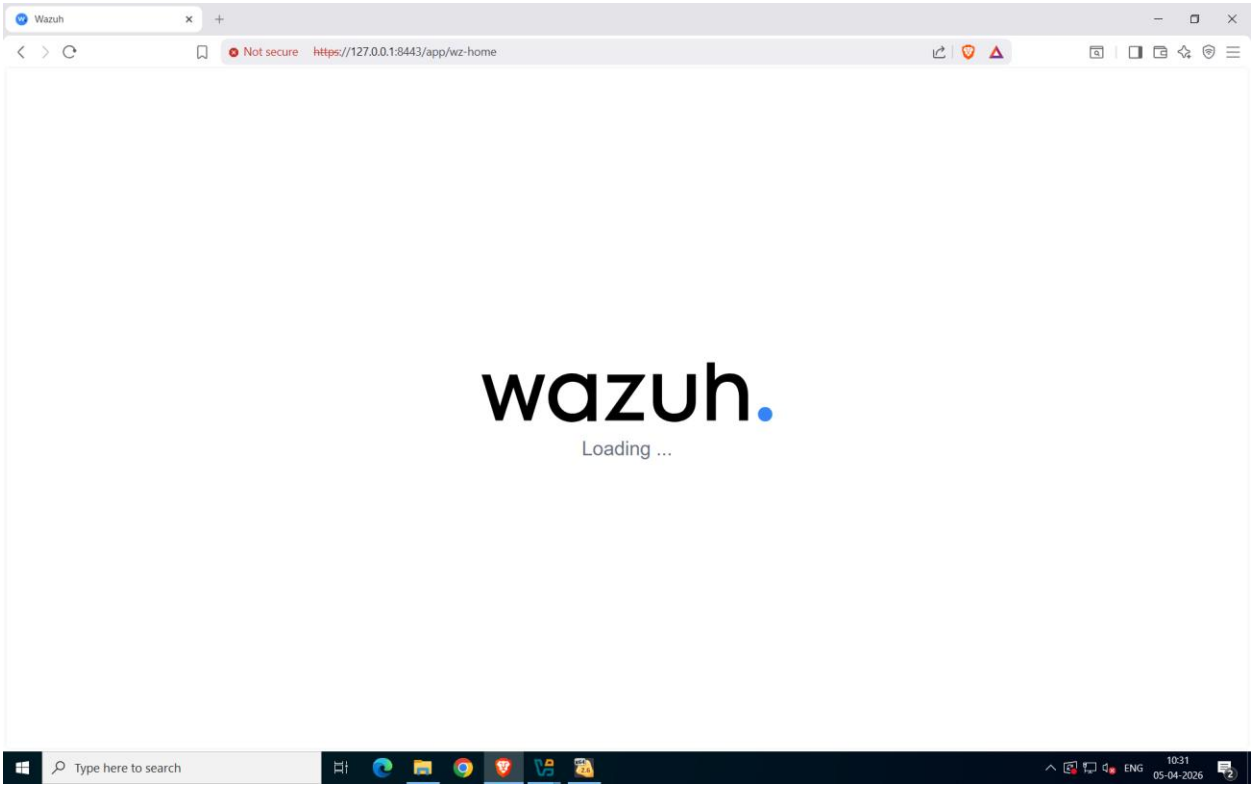

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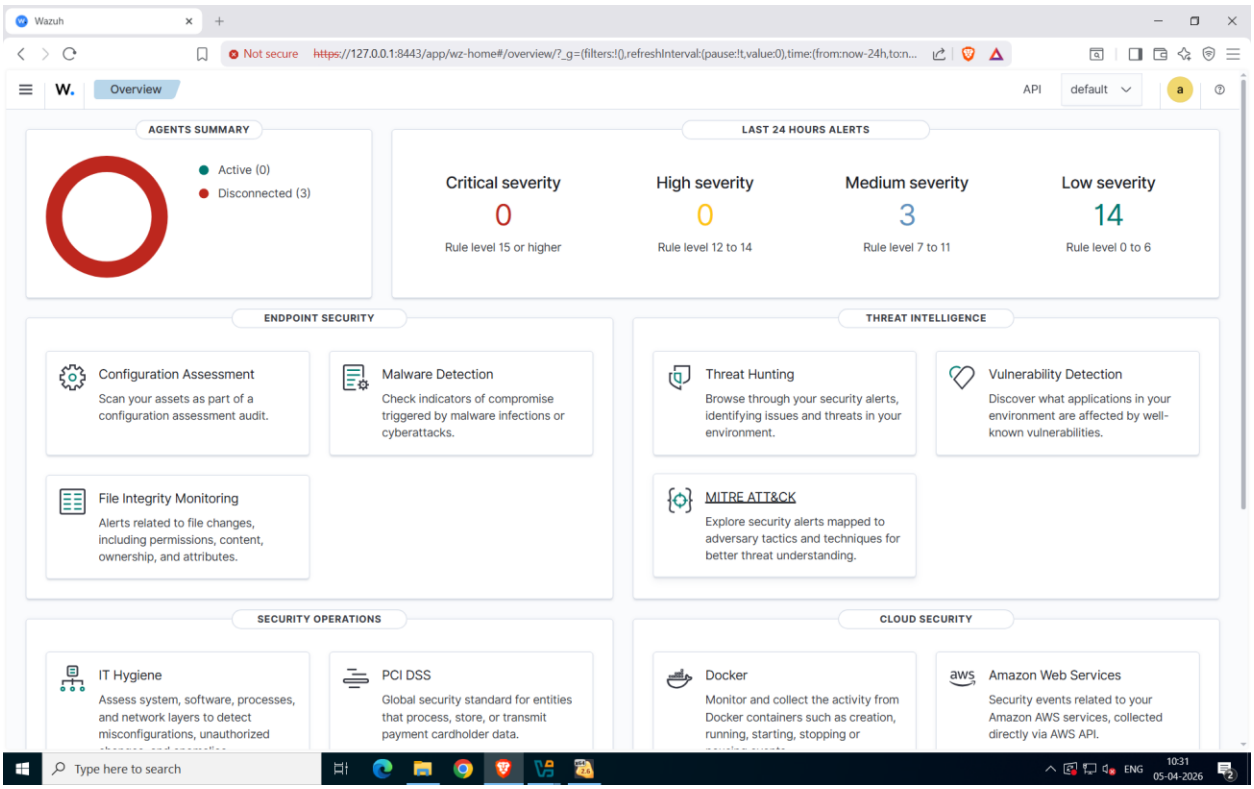


After login the Wazuh Dashboard will be loaded and there will be not agents initially.

5. Wazuh Agent Installation

5.1 Objective

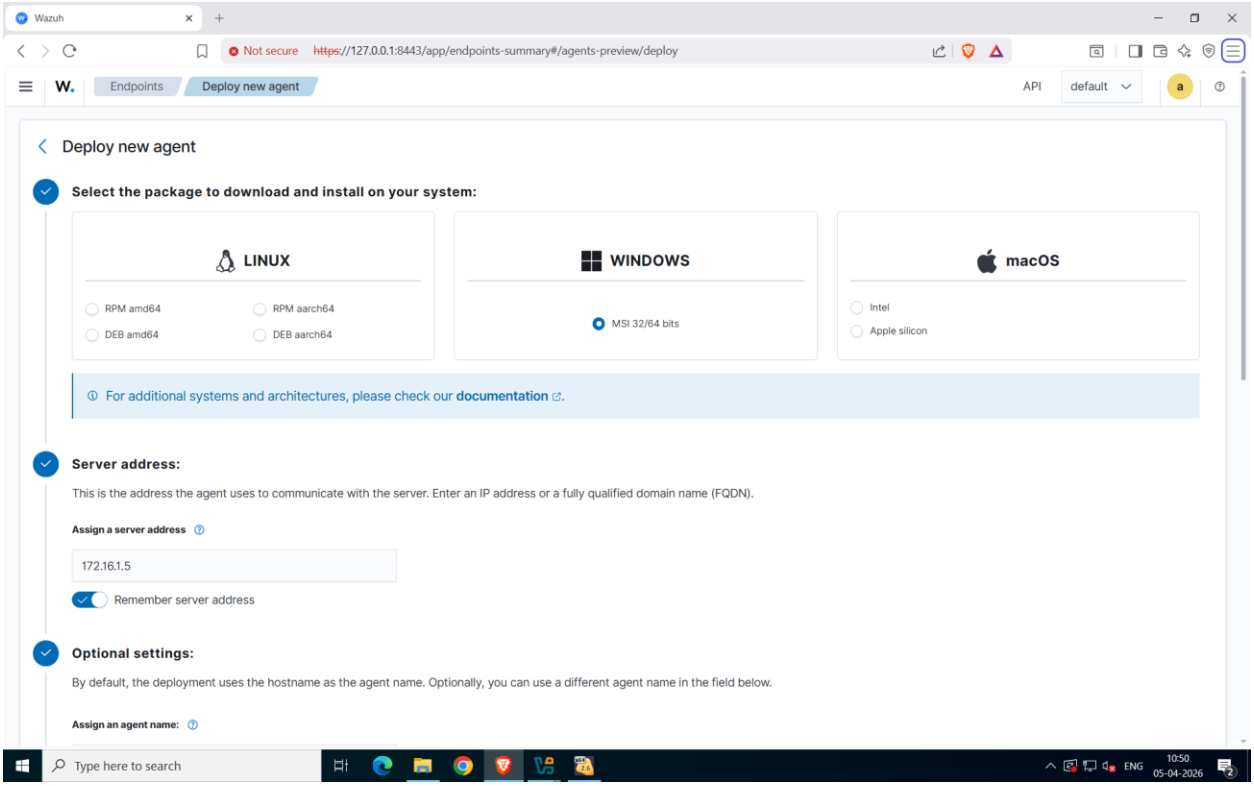
Install the Wazuh agents on all the endpoints to enable centralized monitoring.

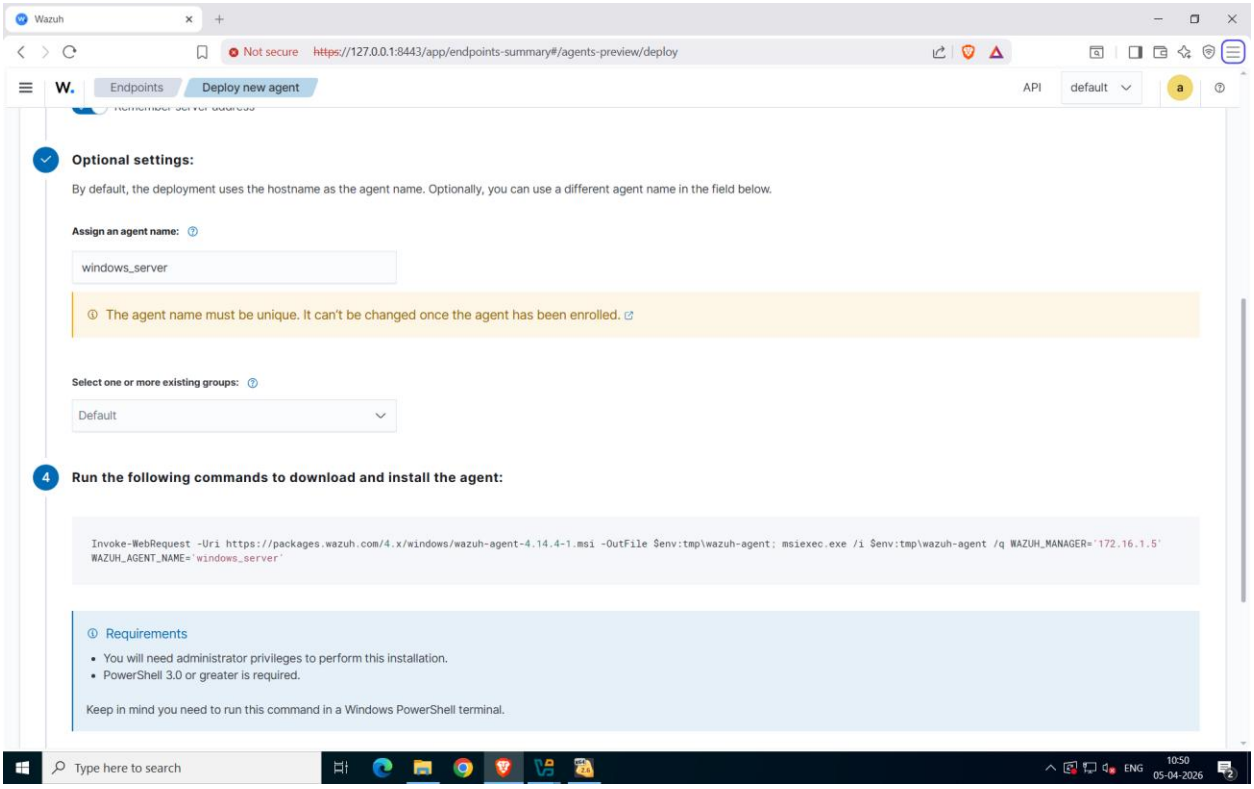
5.2 Deploying Wazuh Agent on Domain Controller (Windows Server)

Steps Performed:

1. Login to Wazuh Dashboard
2. Click on “Deploy new agent”
3. Select windows MSI 32/64 bits
4. Enter Wazuh VM IP: 172.16.1.5 and also click remember server address
5. Enter agent name (optional)
6. Copy the commands given on the dashboard
7. Go to your windows server -> open powershell and run it as administrator
8. Execute the commands given on the dashboard into the powershell
9. Restart windows server
10. Return to the Wazuh dashboard and click “back to agent list”

Now the Wazuh agent is installed on the Domain Controller (Windows Server).





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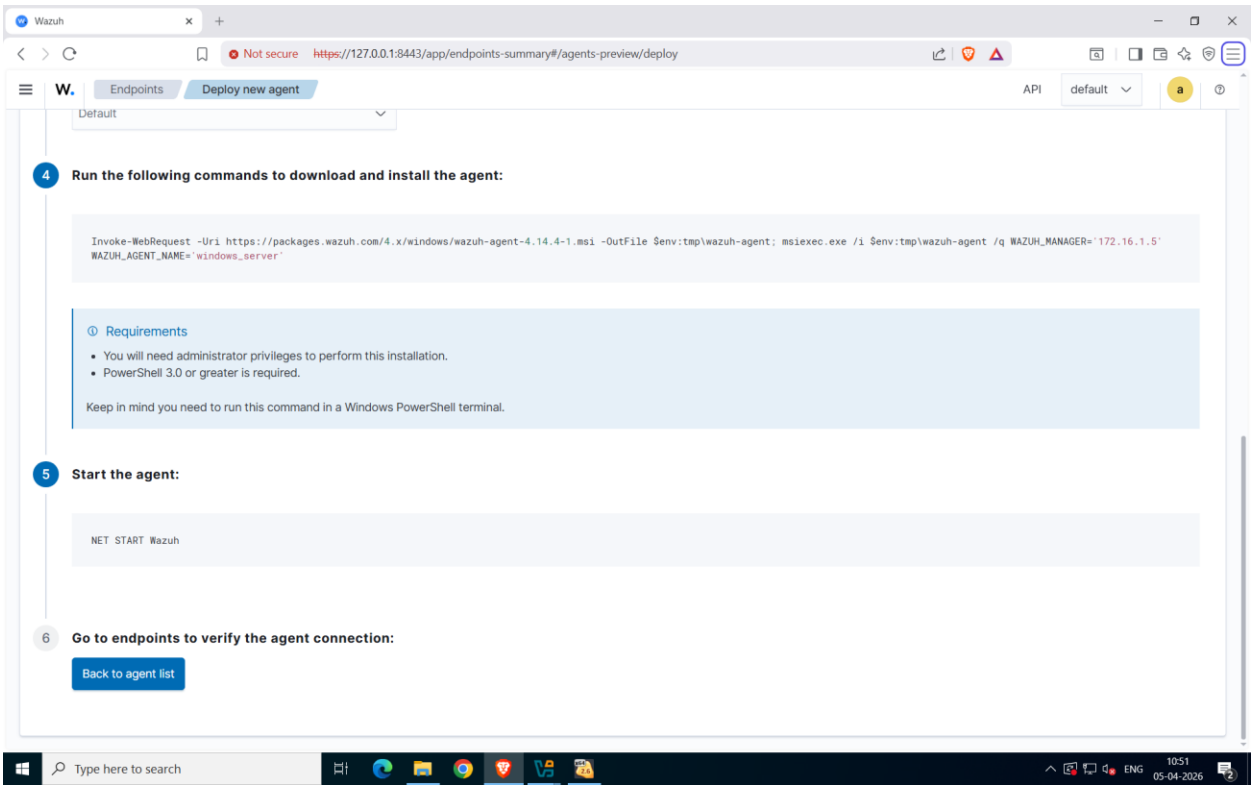

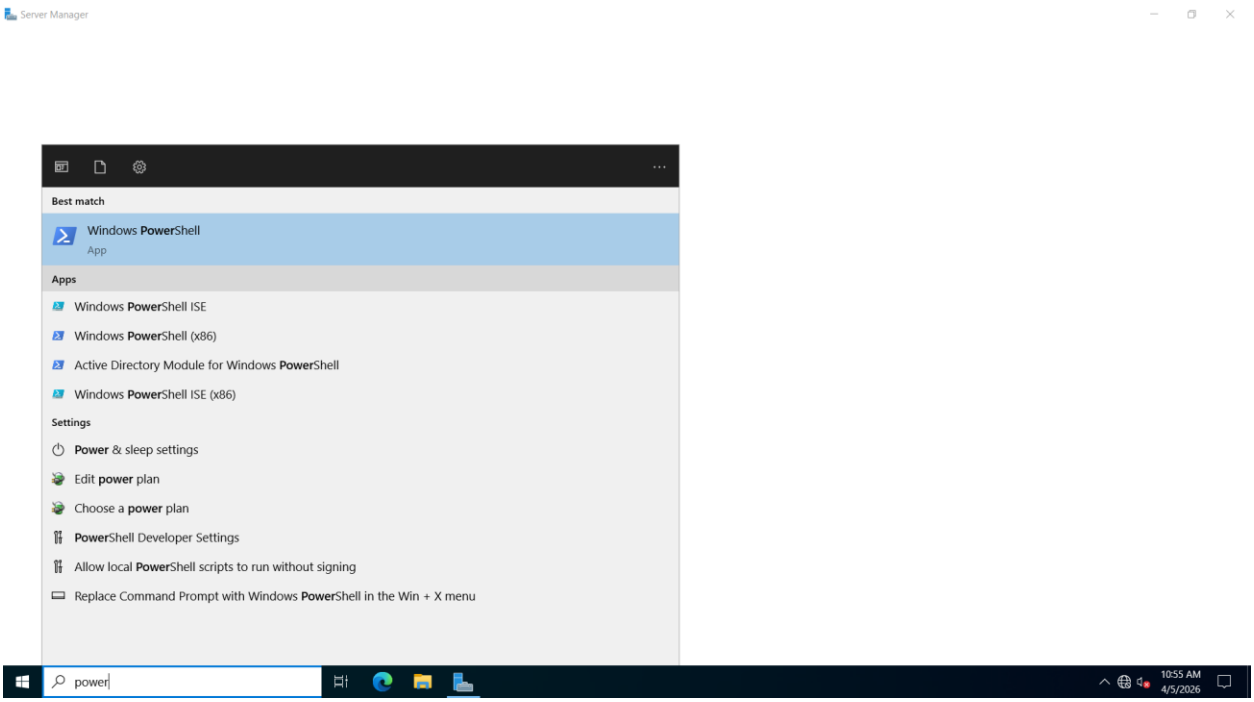

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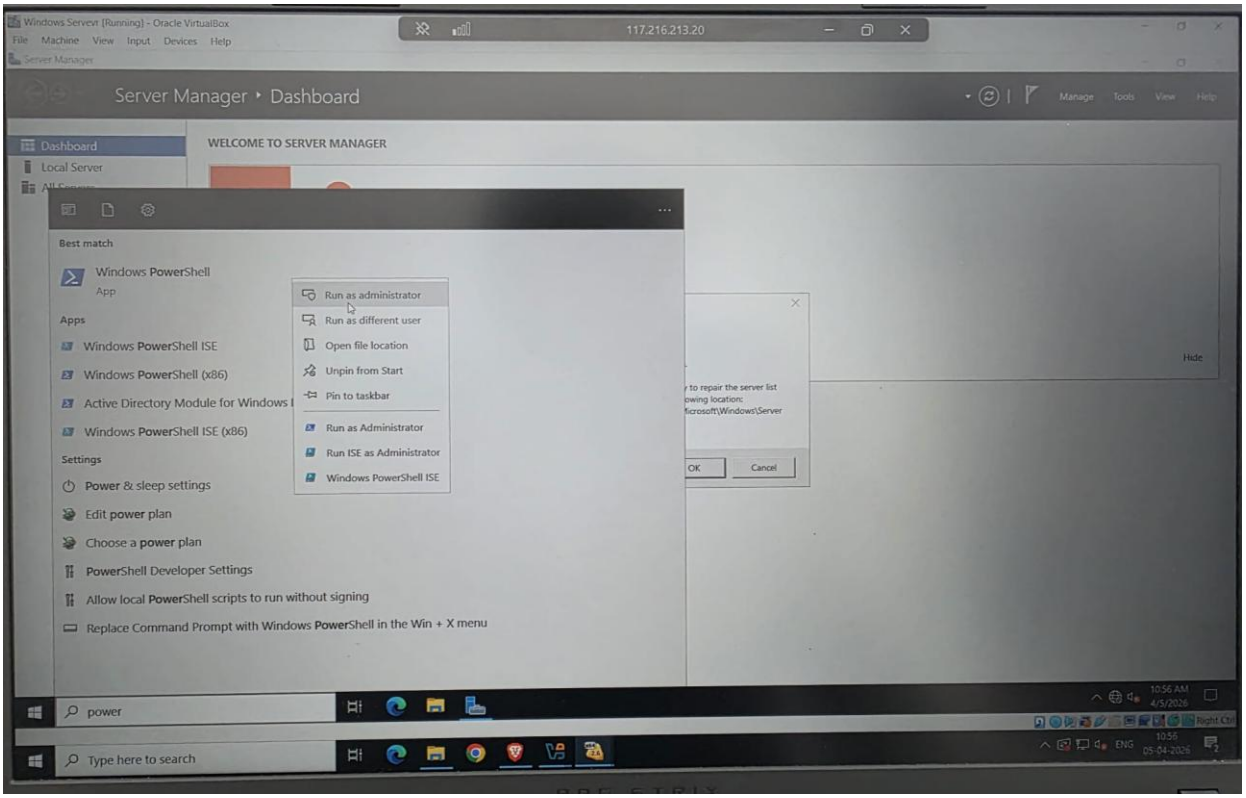

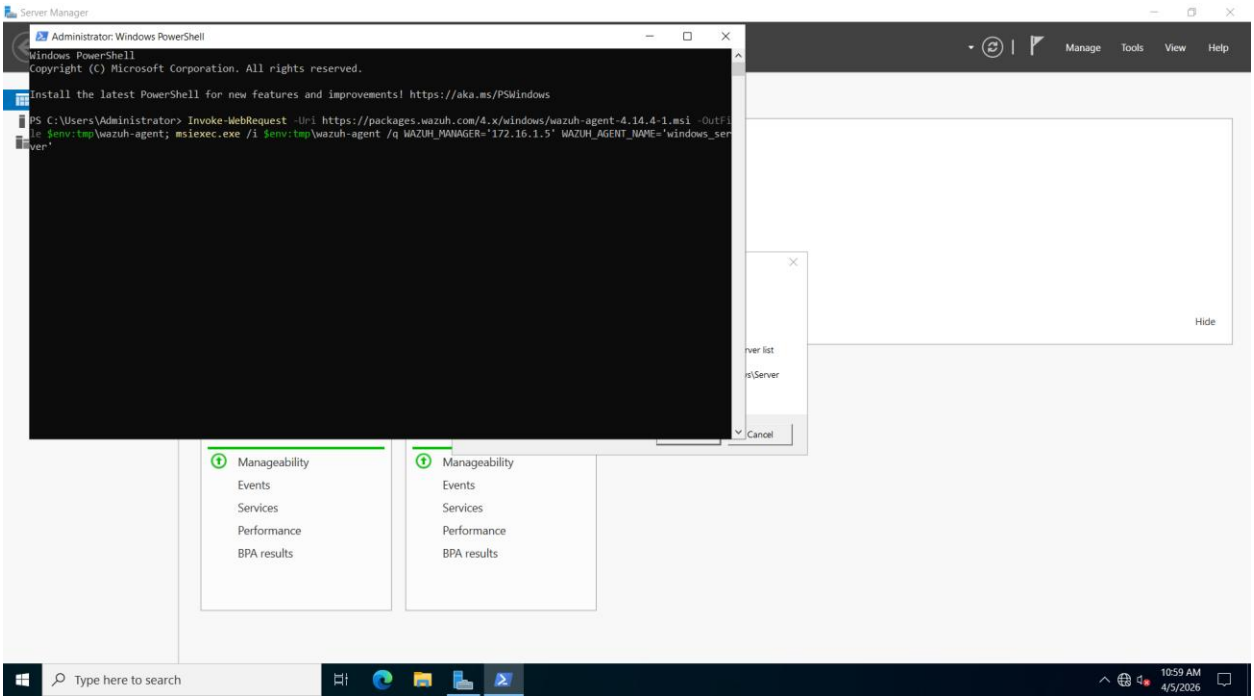

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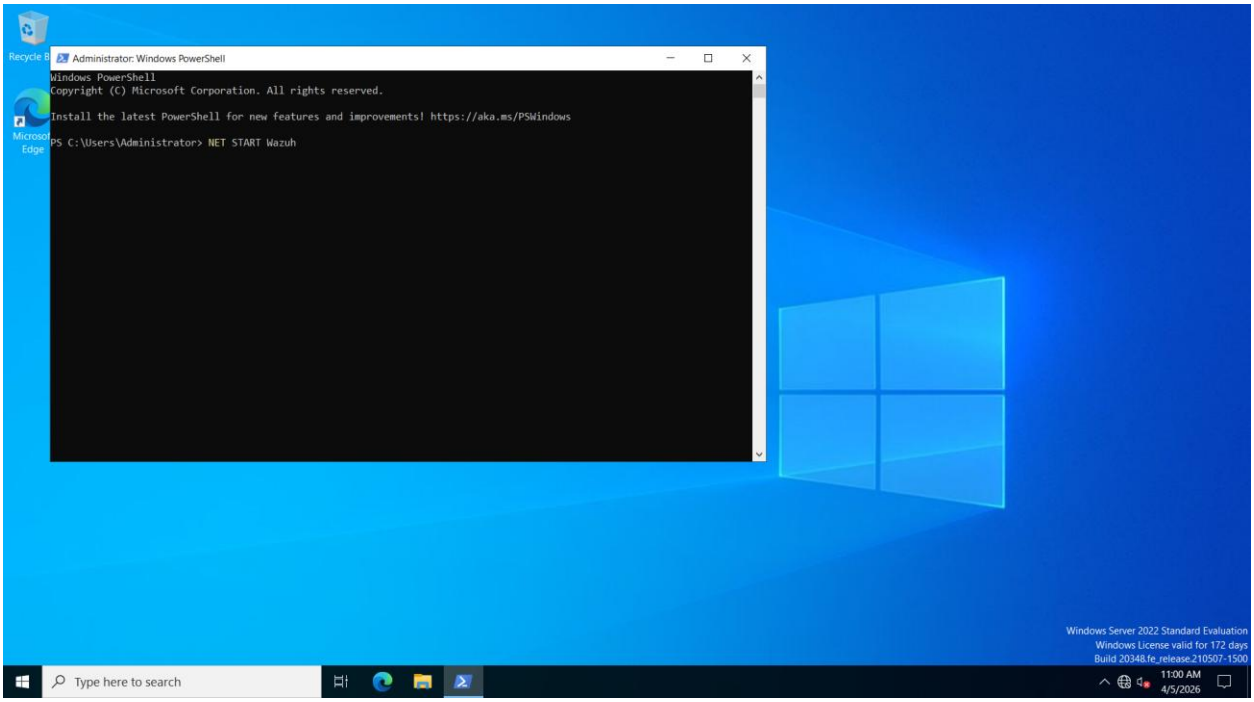

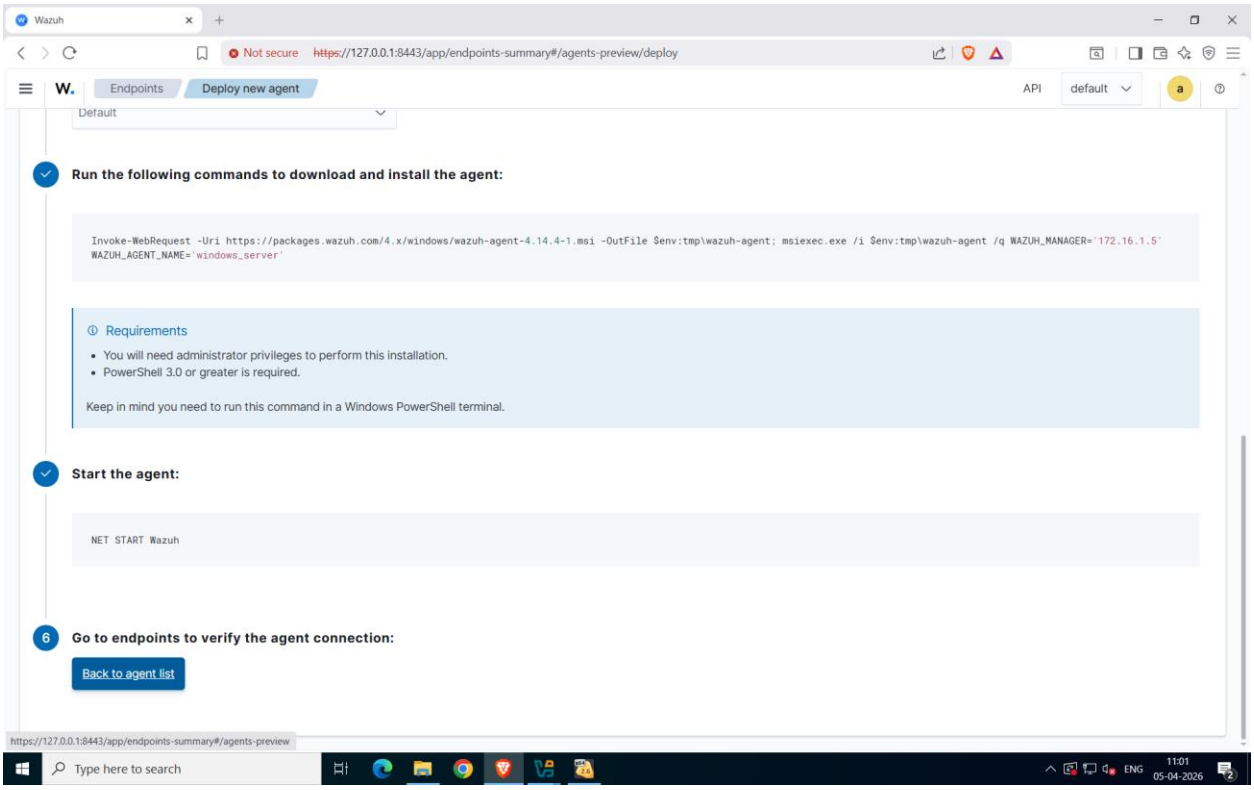

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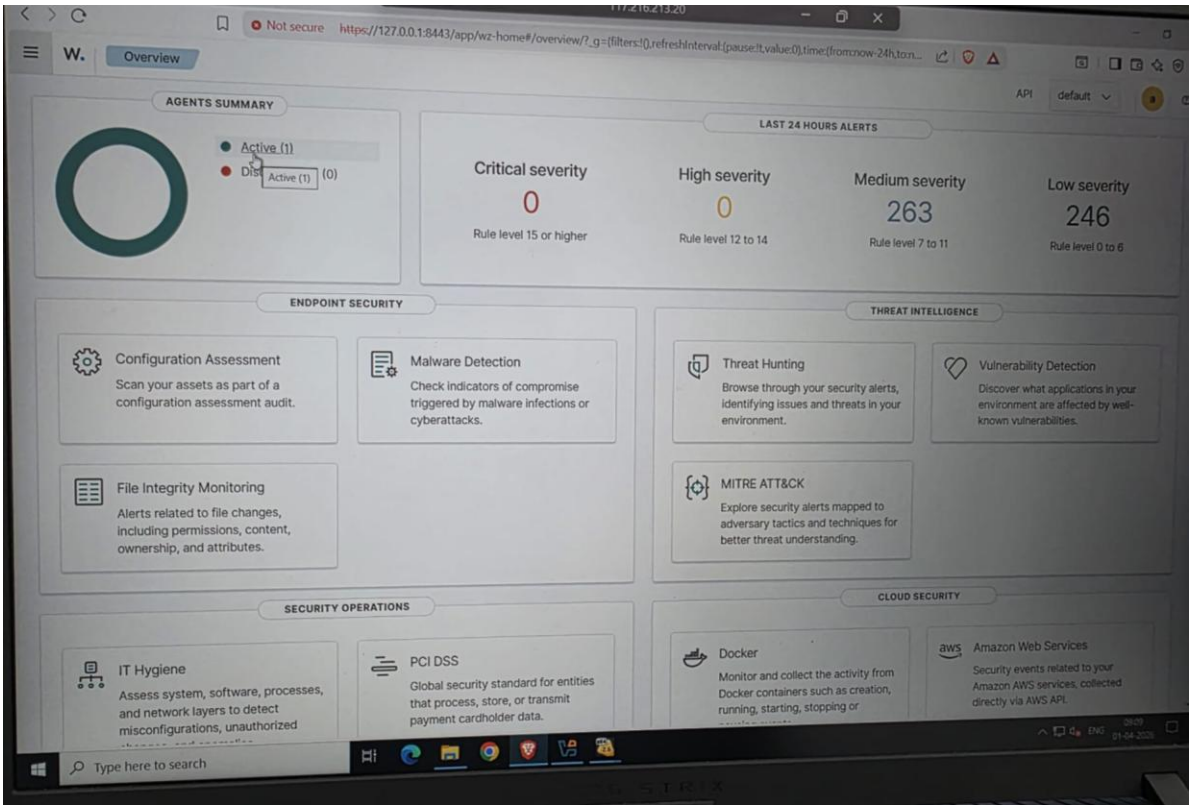

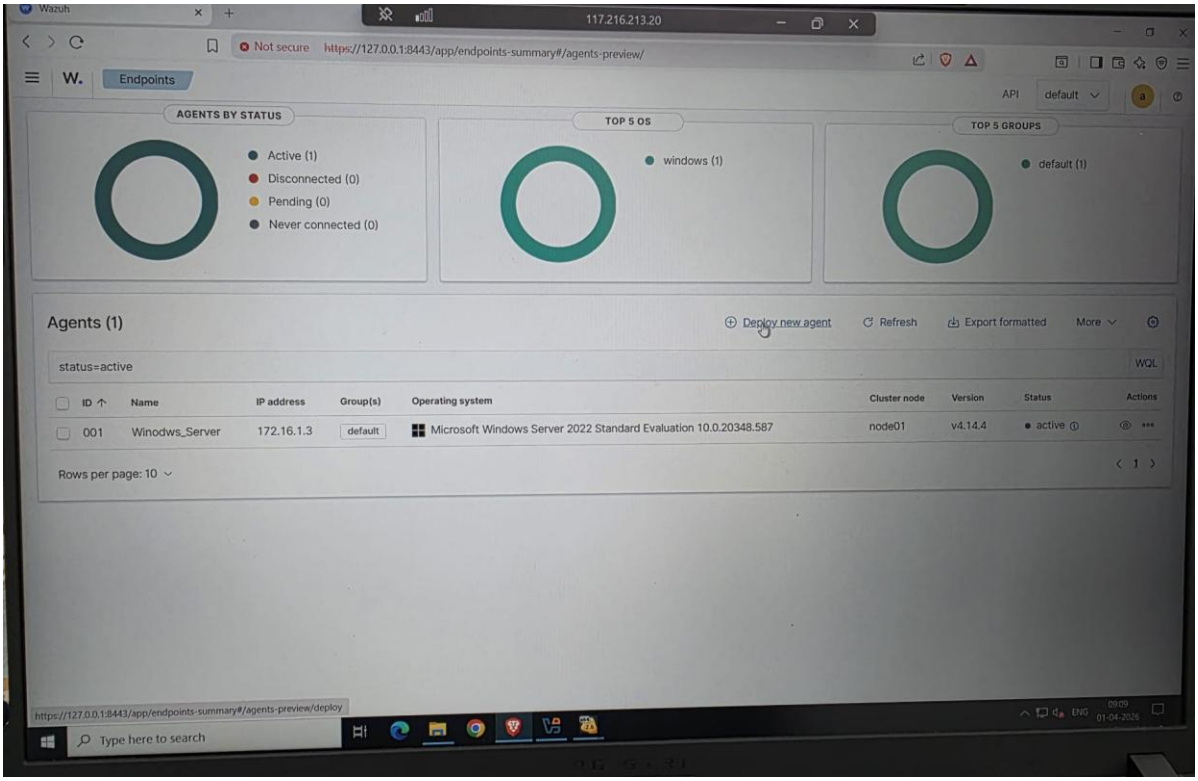

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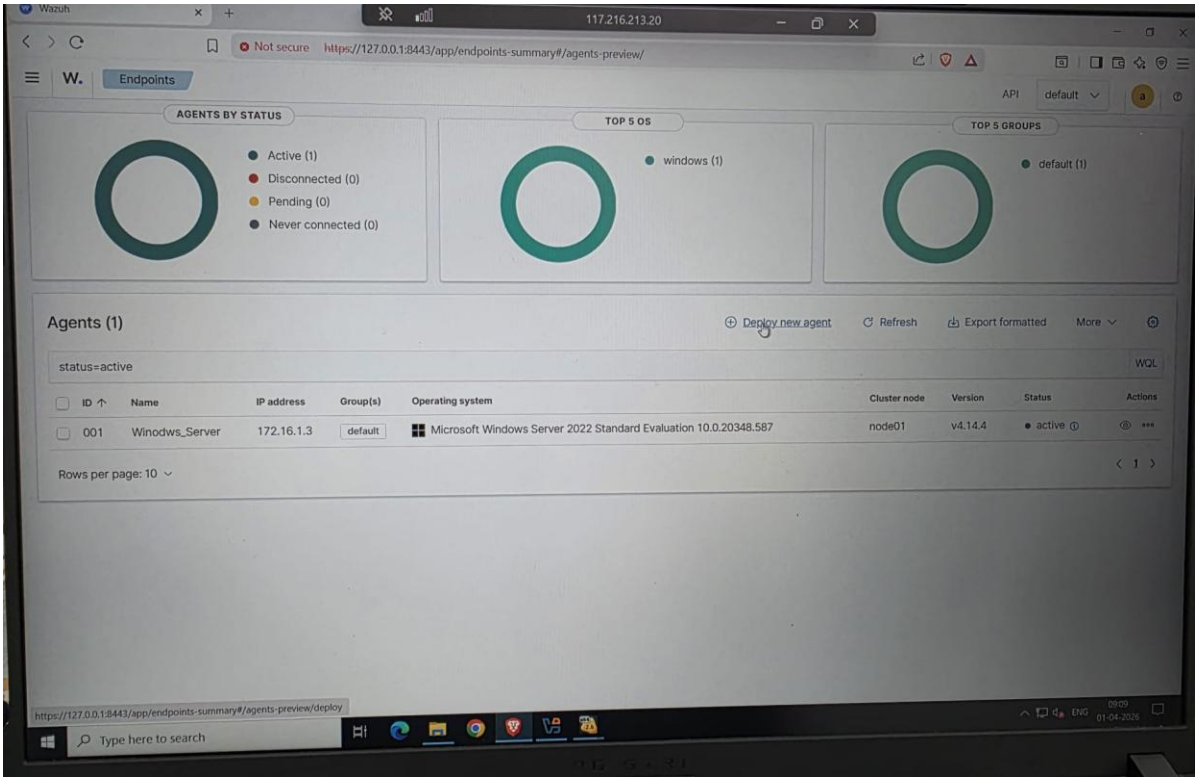
5.3 Deploying Wazuh Agent on Windows 10 Client

Steps performed:

1. Click “Deploy new agent”
2. Select “MSI 32/64 bits”
3. Enter agent name (optional)
4. Start windows 10 client and run powershell as administrator
5. Execute the commands in powershell given in Wazuh dashboard
6. Return to the Wazuh dashboard and click “back to agent list”



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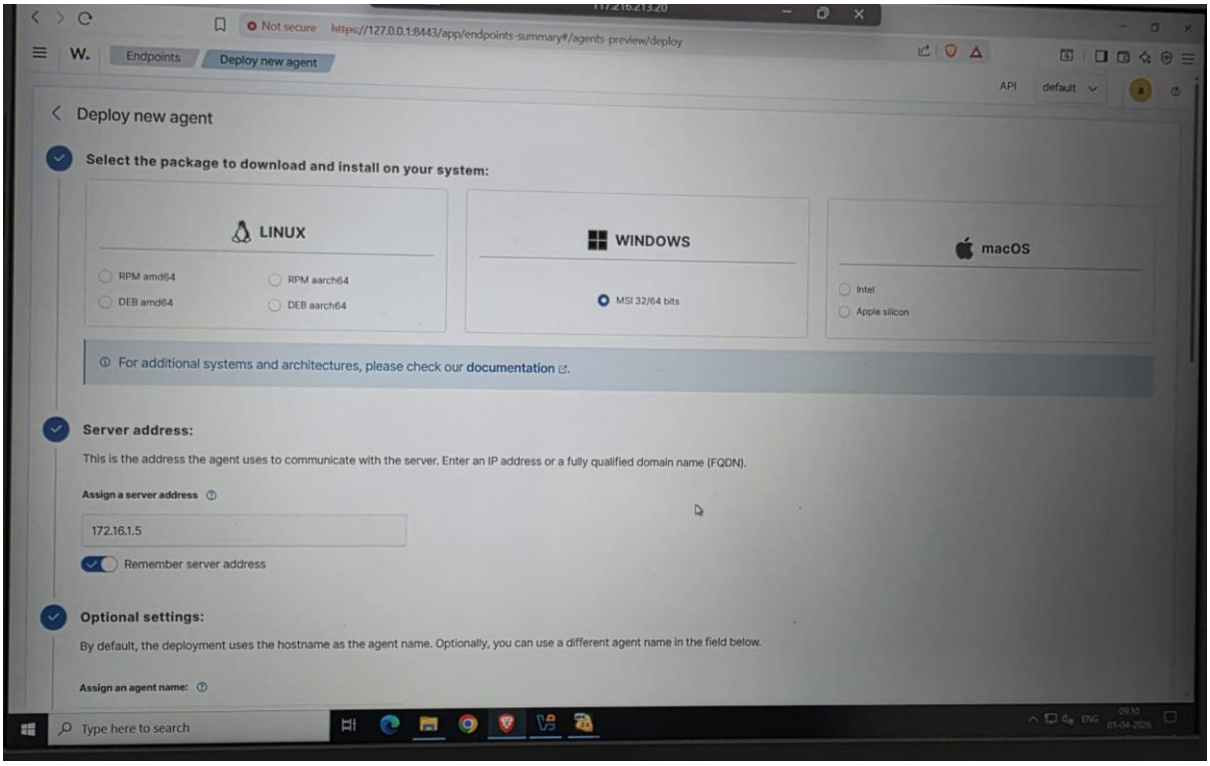

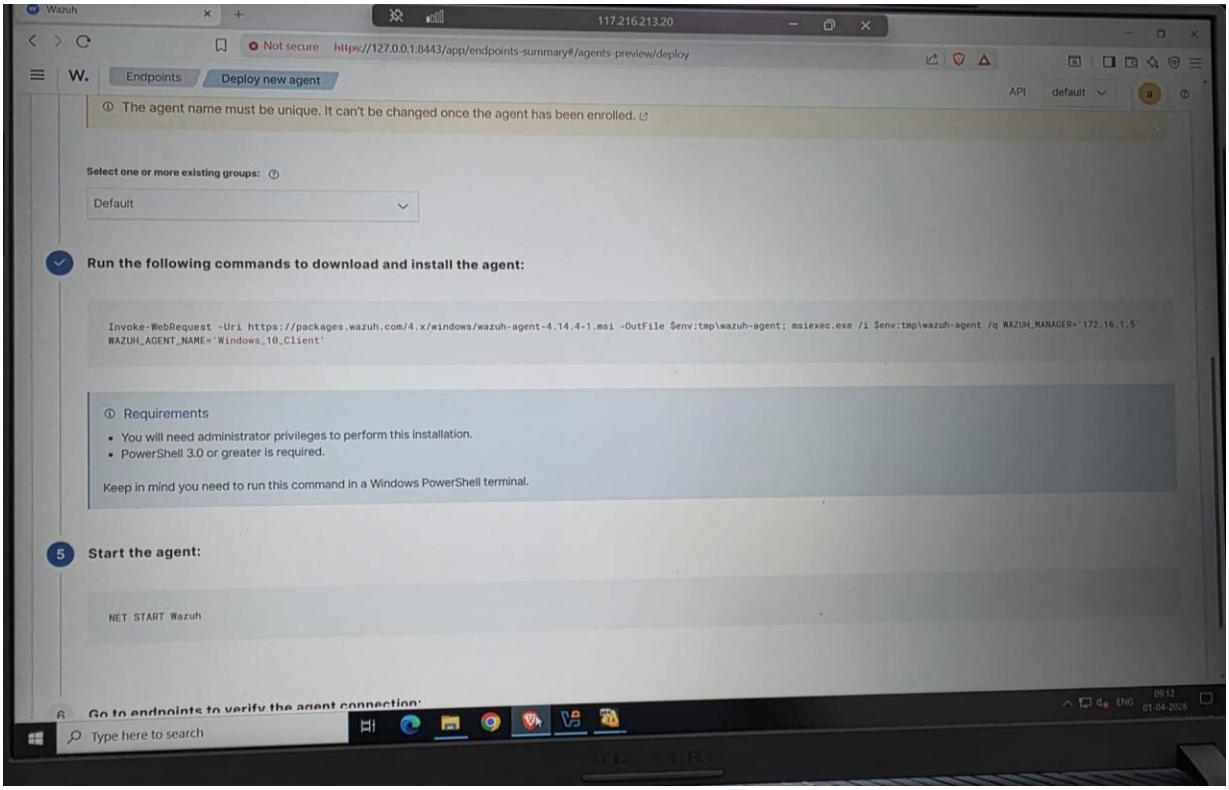

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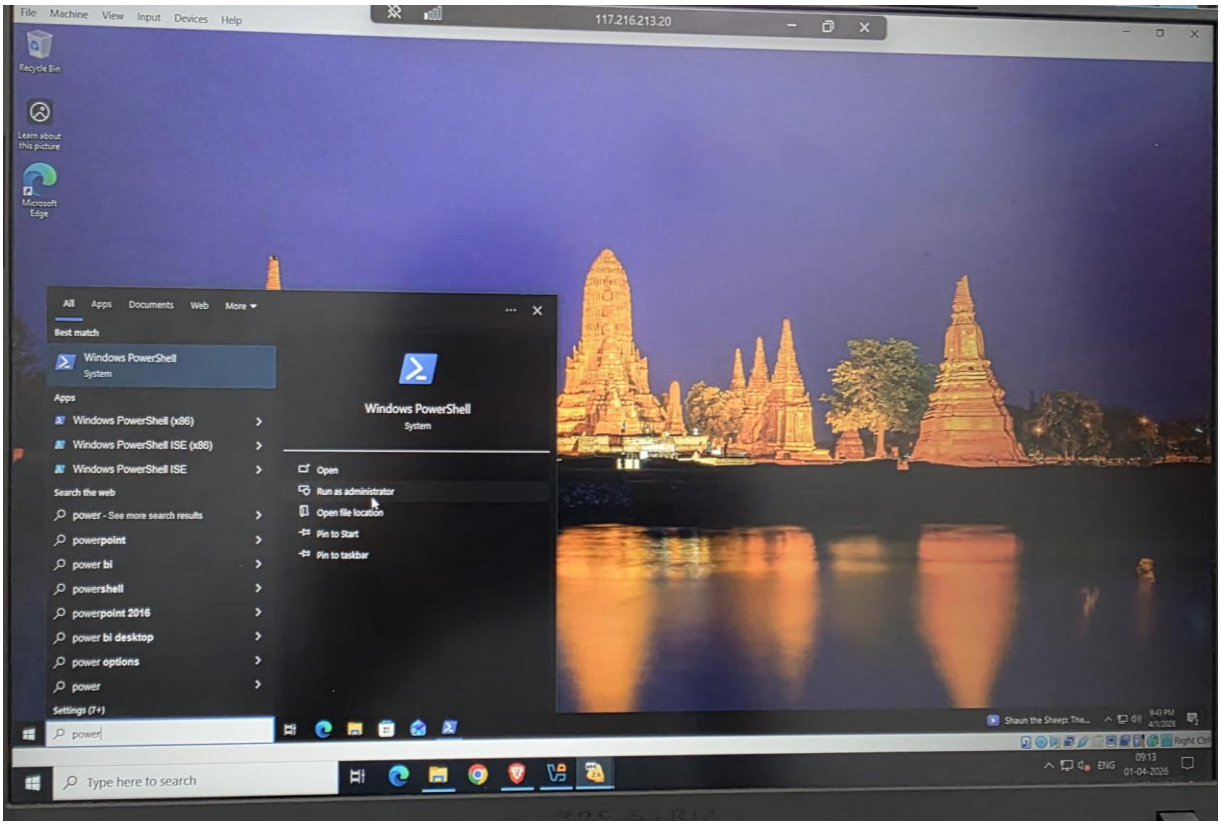

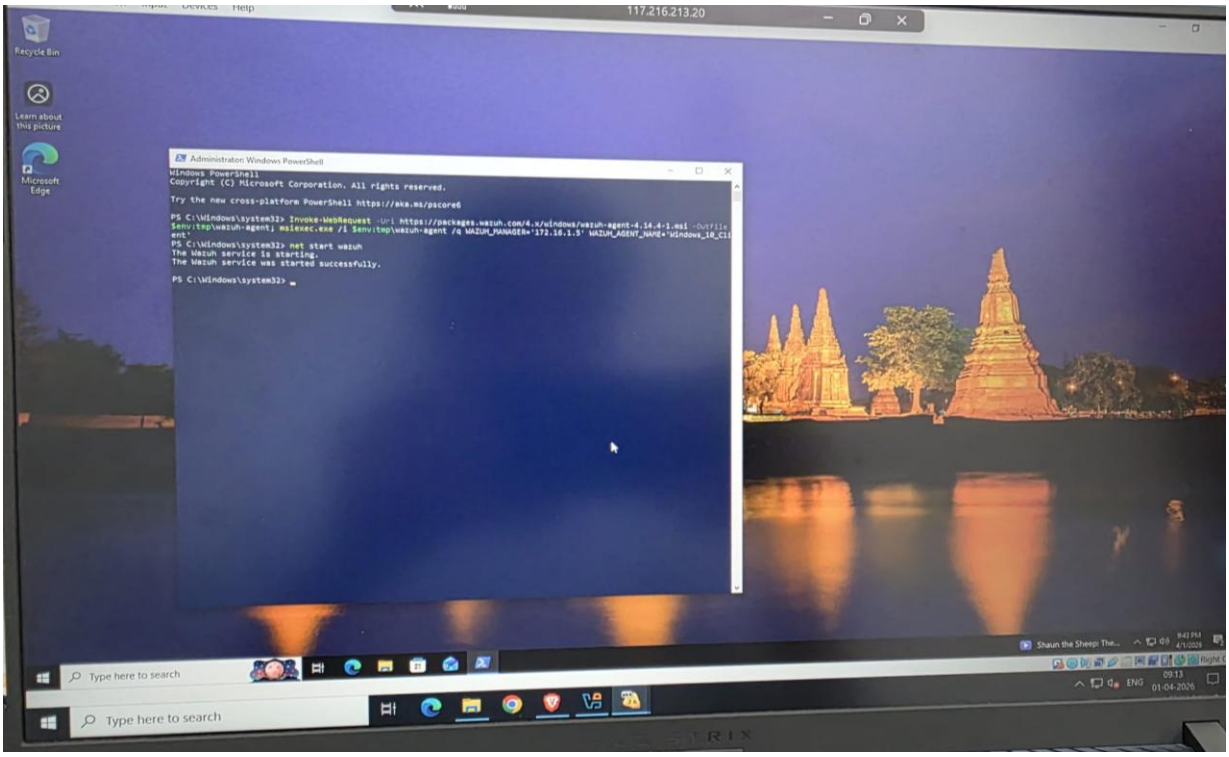

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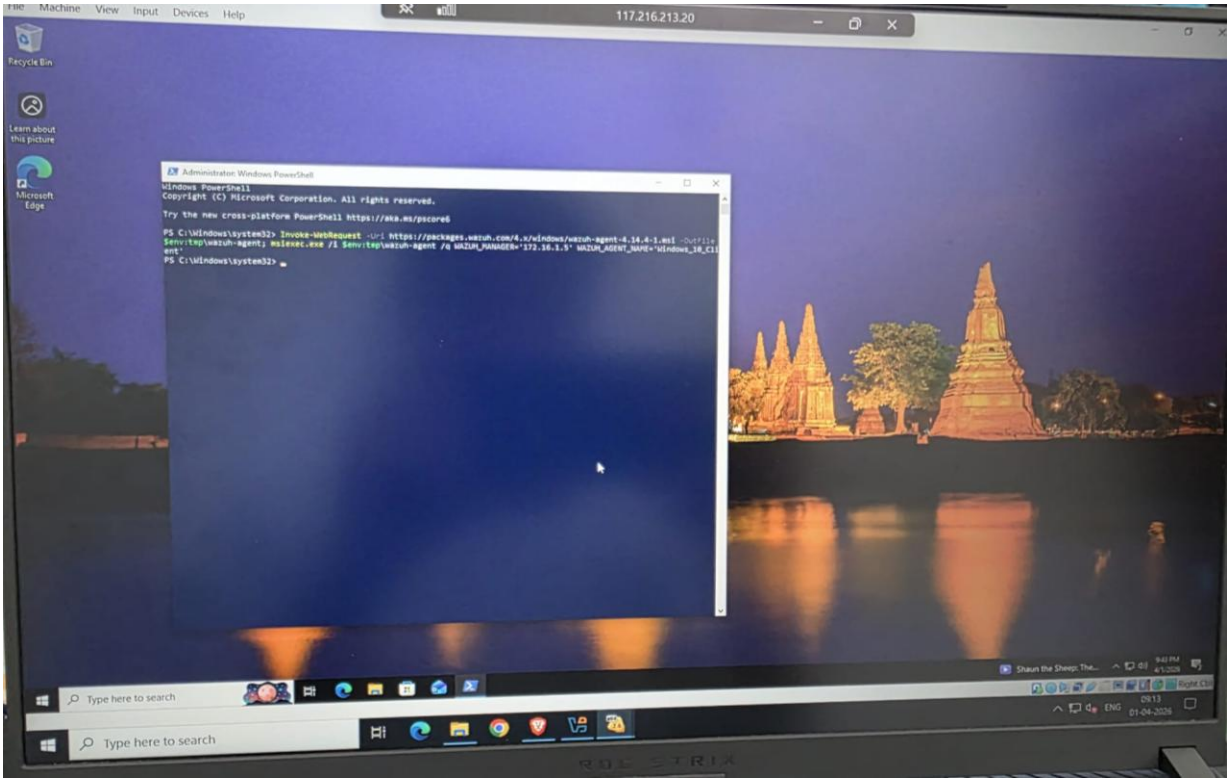

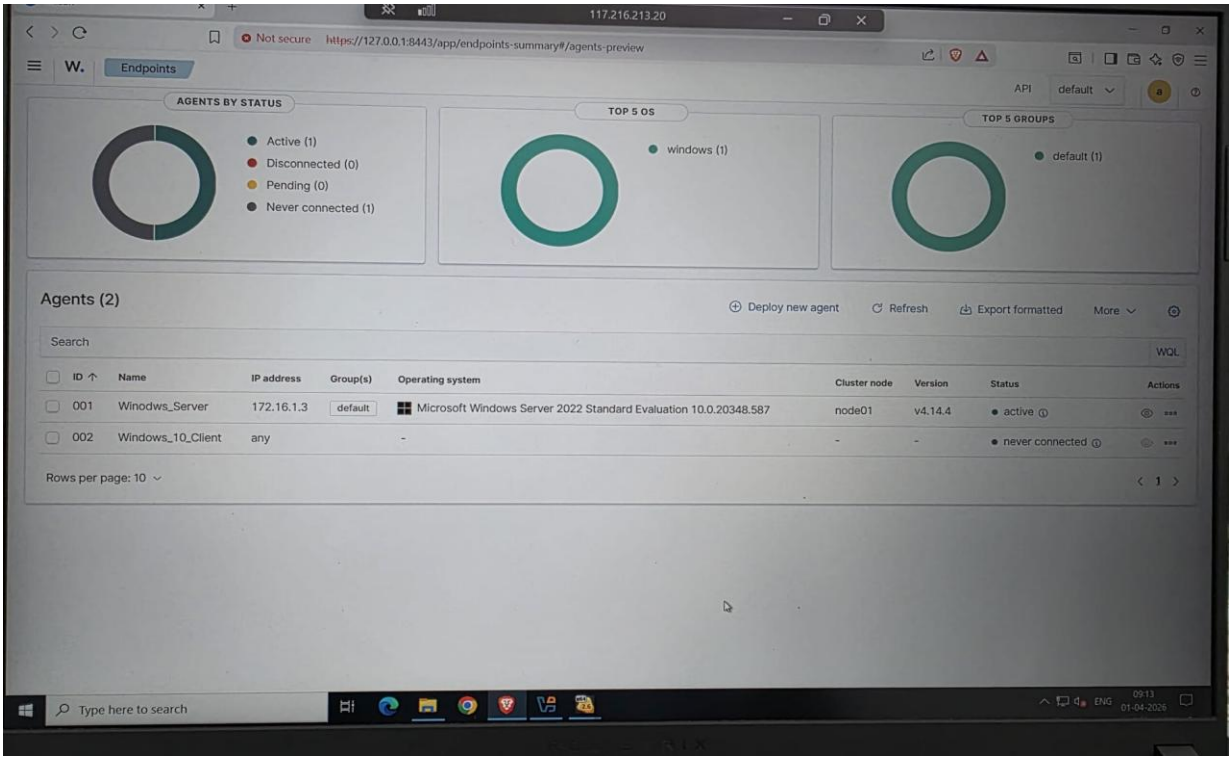

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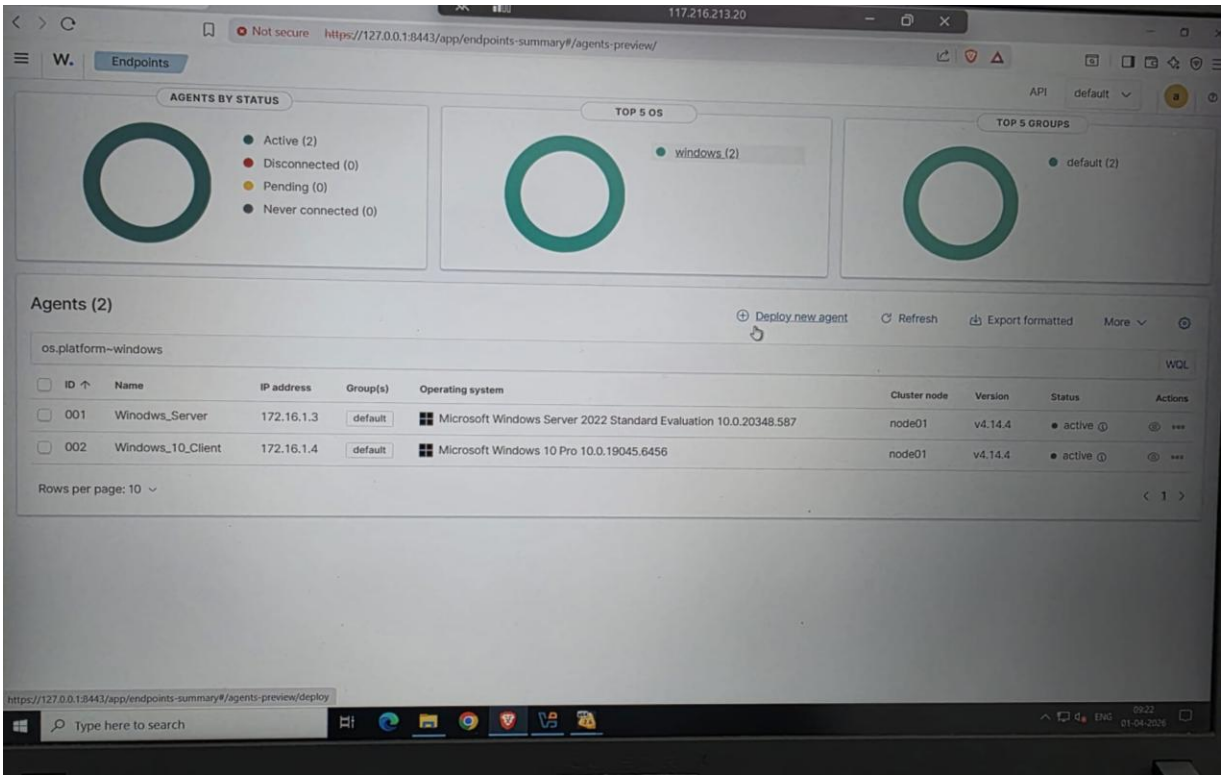

5.4 Deploying Wazuh Agent on Ubuntu Client

Steps performed:

1. Click “Deploy new agent”
2. Select “DEB amd64”
3. Enter agent name (optional)
4. Start ubuntu client and open terminal and switch to root user
5. Execute the commands in terminal given in Wazuh dashboard
6. Return to the Wazuh dashboard and click “back to agent list”

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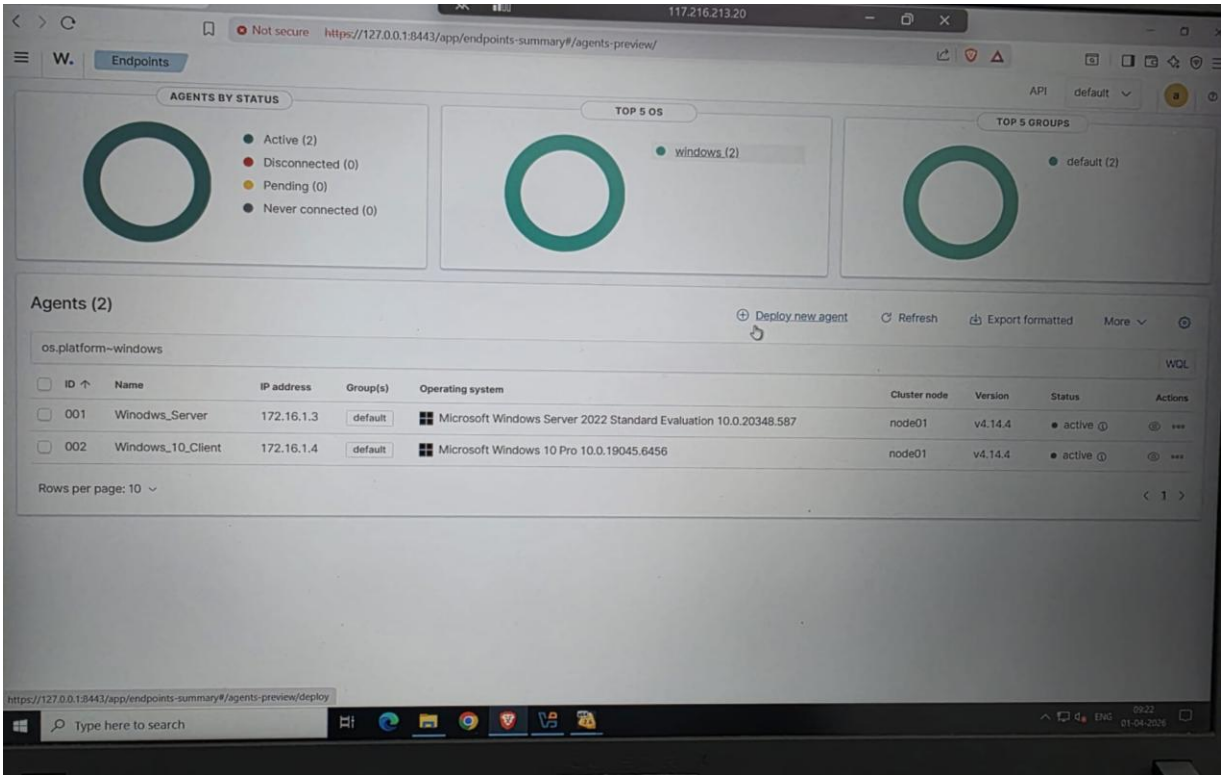

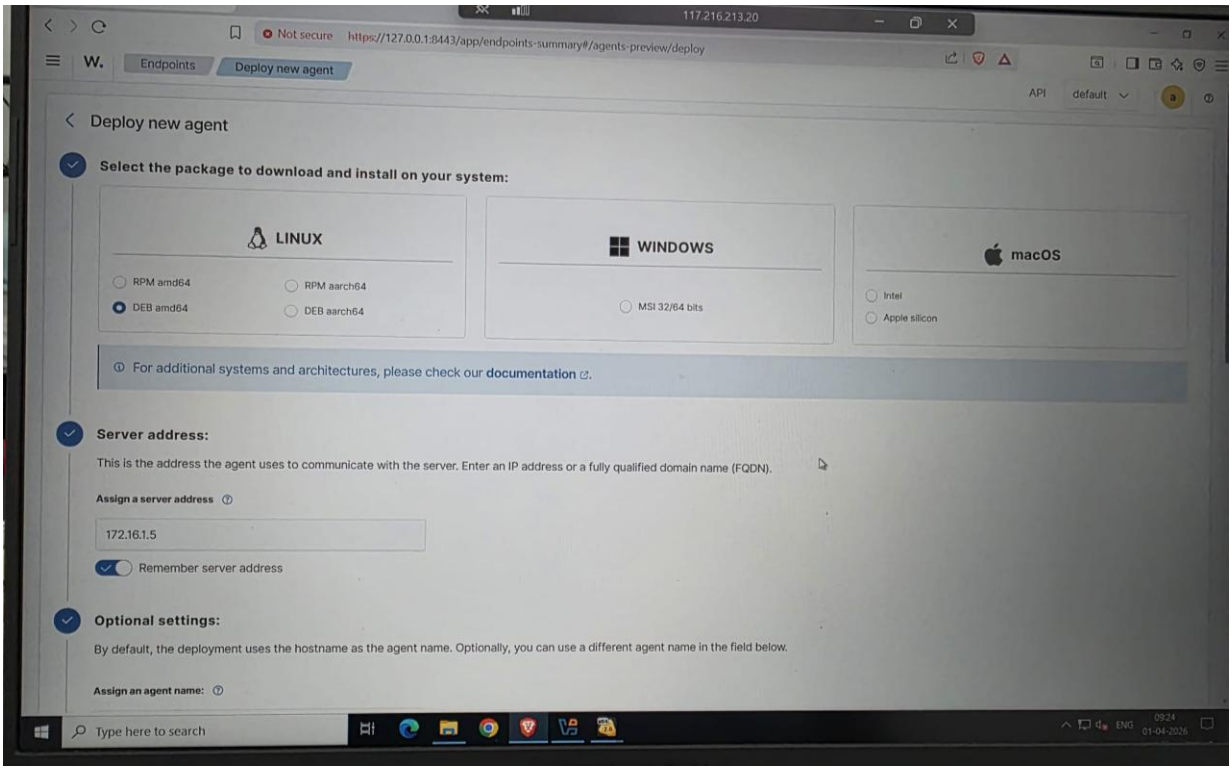

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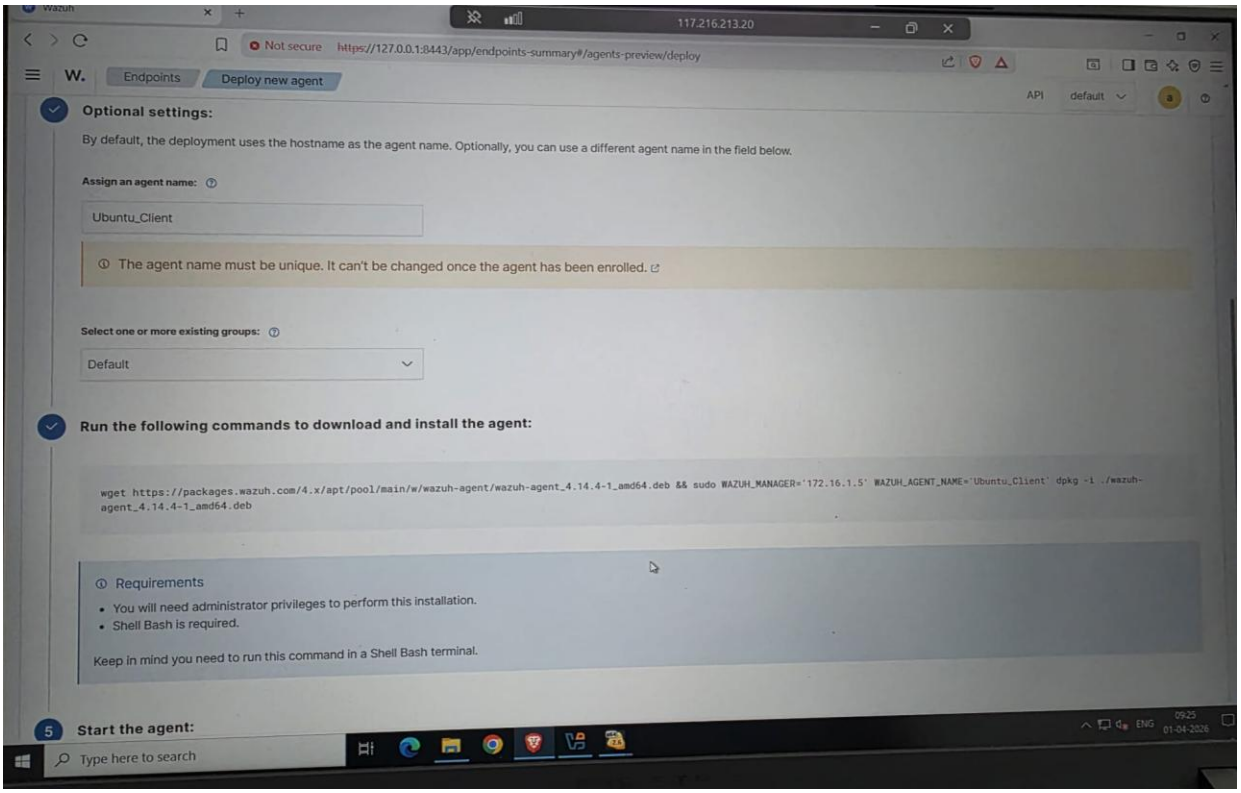

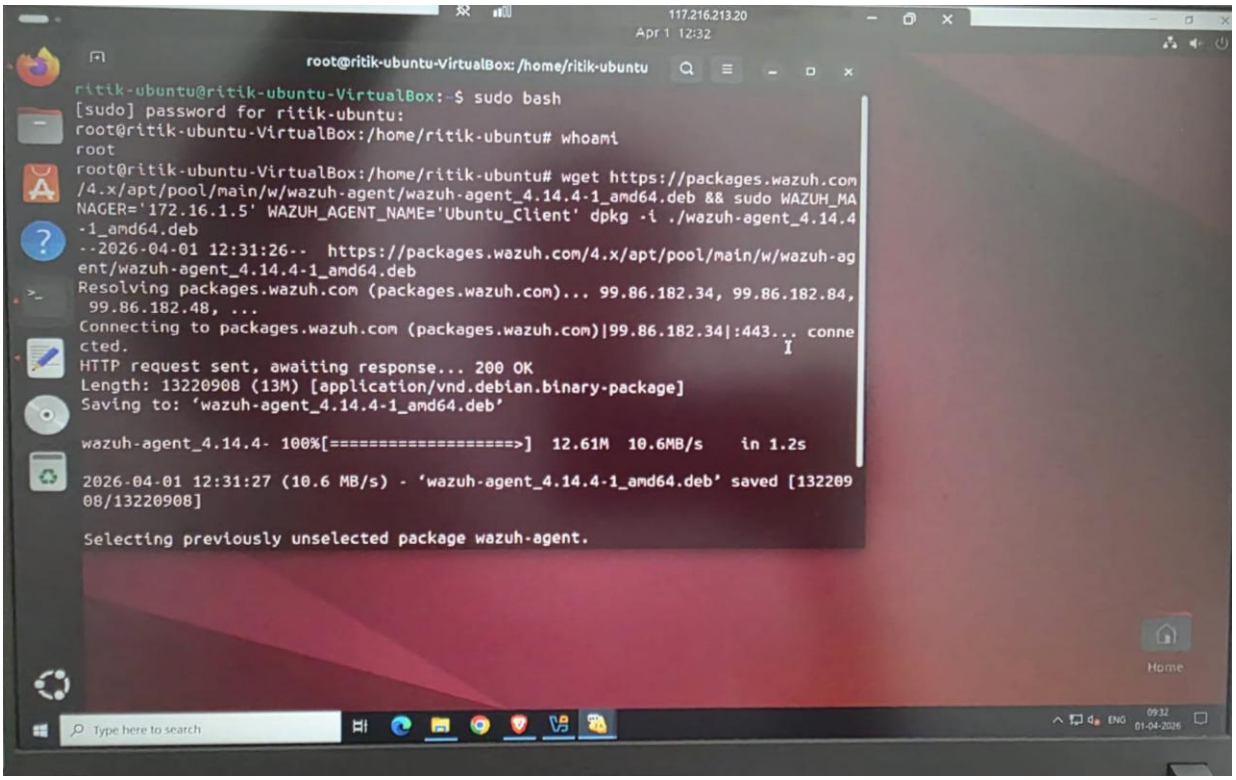

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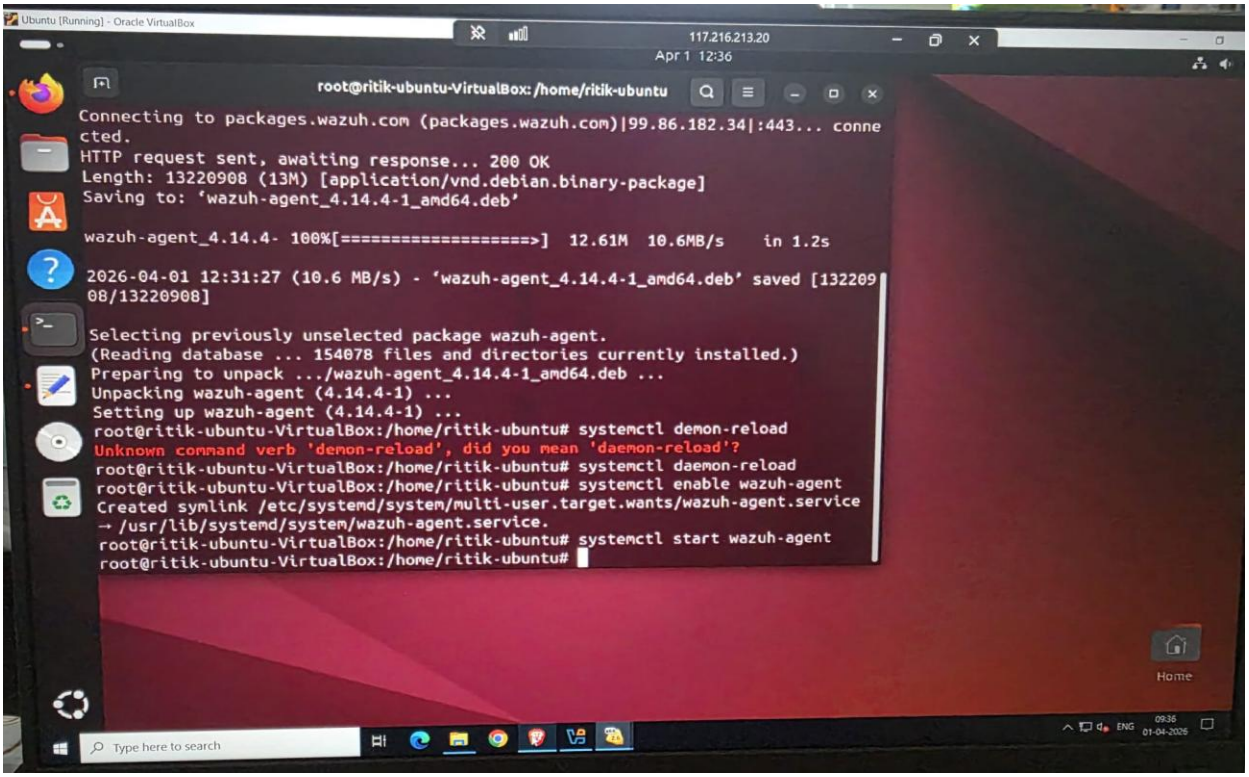

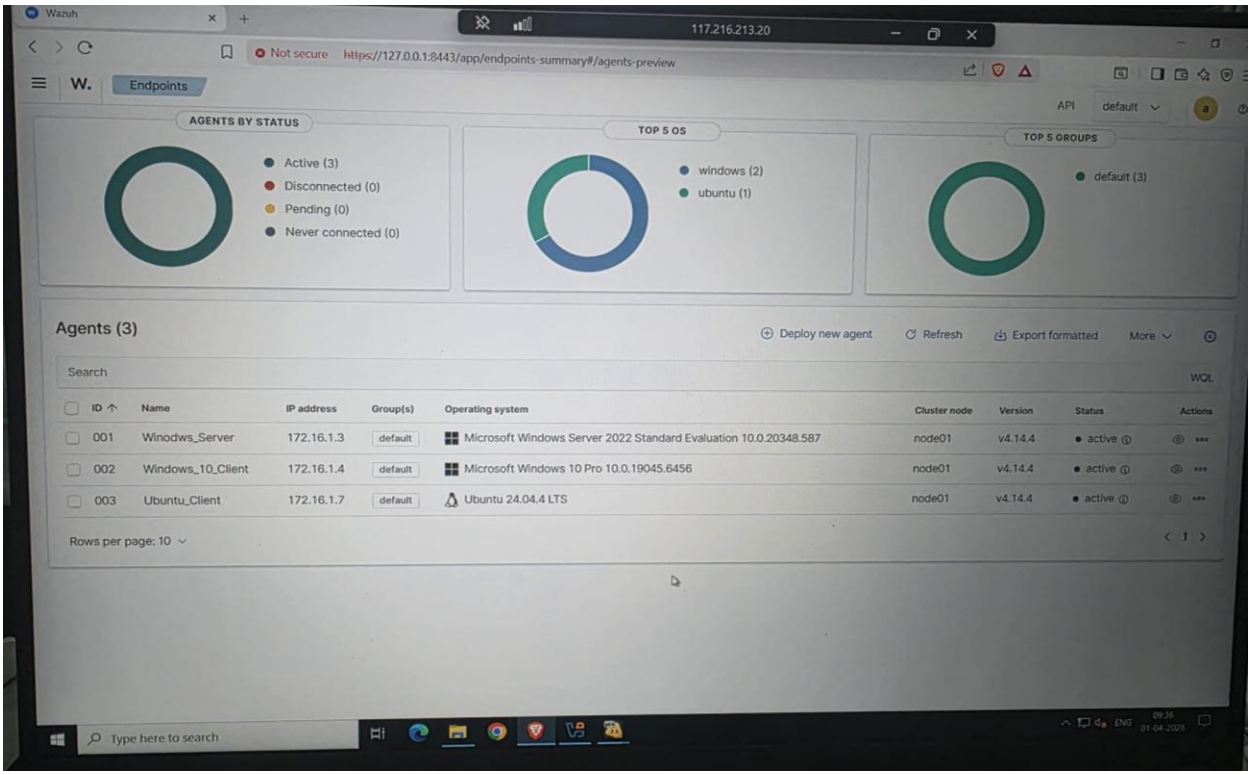

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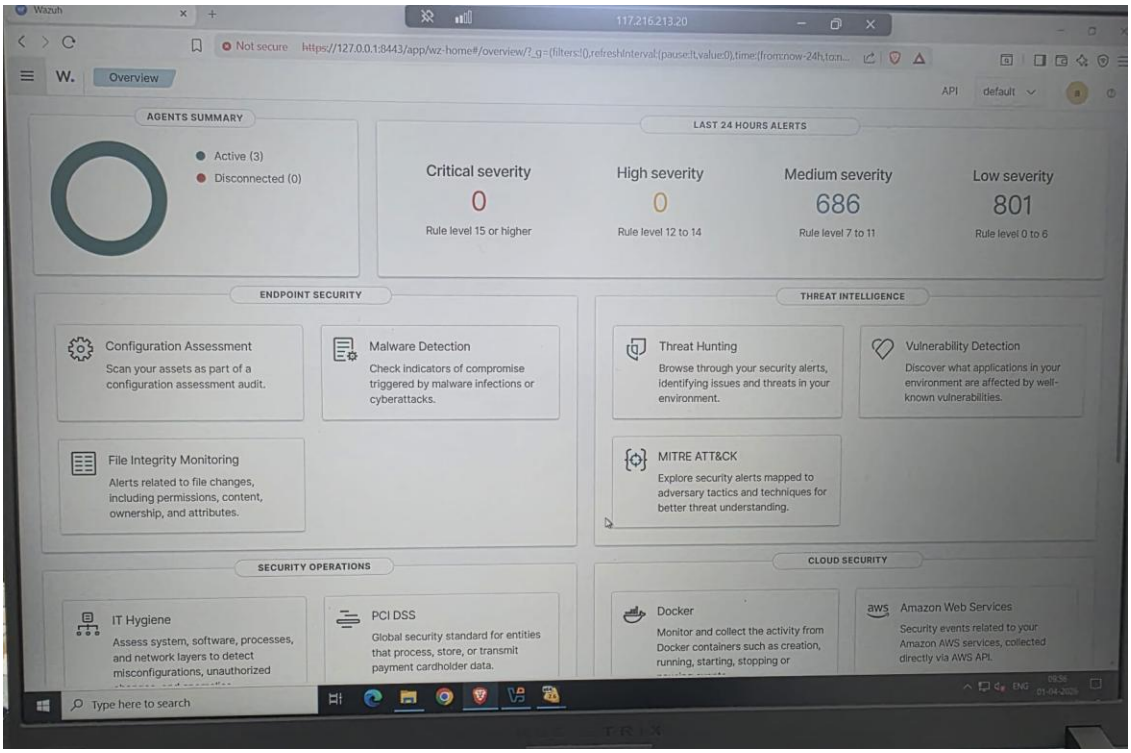

5.5 Verifying Agents Connectivity

To verify that all of the deployed agents are working correctly we can go to the discover tab in Wazuh dashboard and we can see their logs which confirm that all the agents are working correctly.

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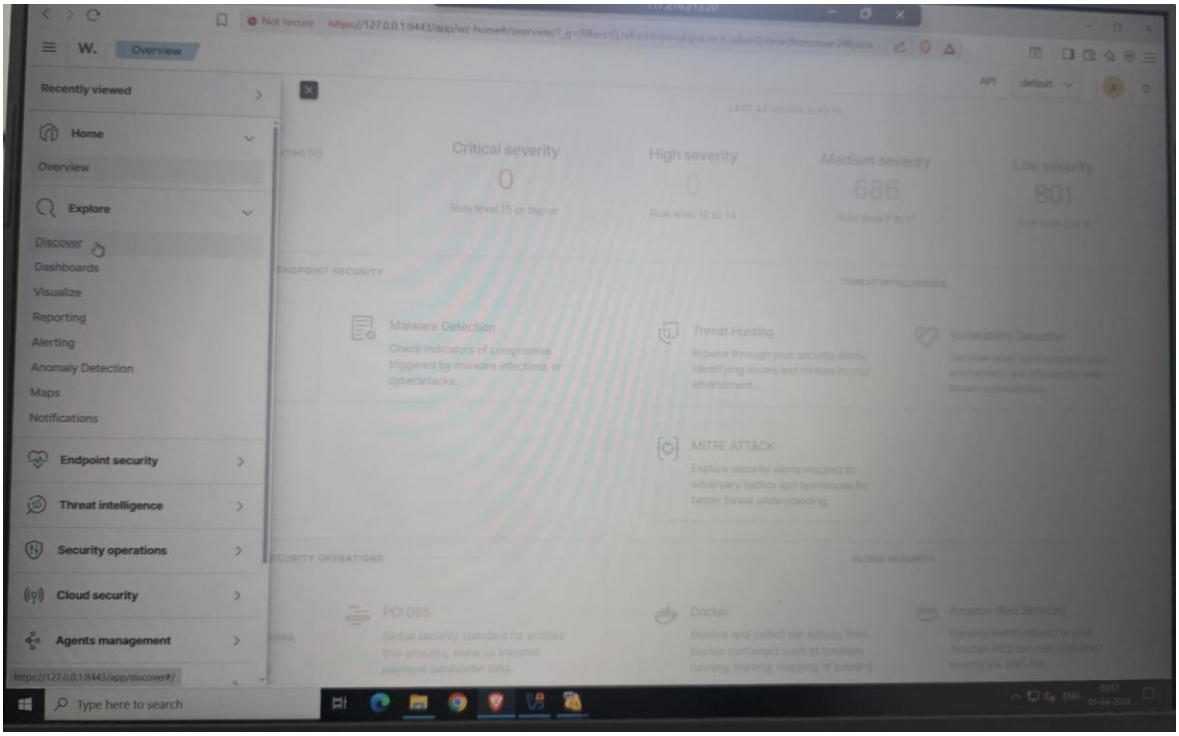

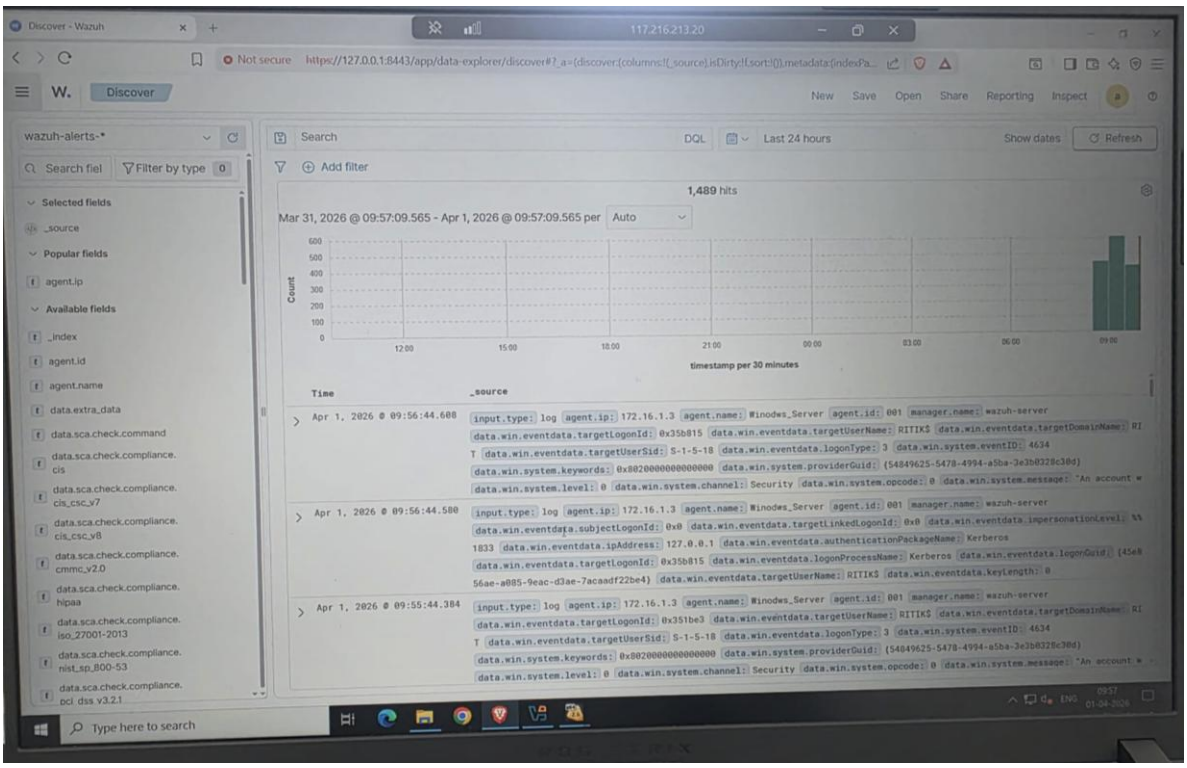

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6. Conclusion

6.1 Summary

In this project, a complete Wazuh SIEM environment was successfully deployed and integrated within an existing Active Directory-based virtual lab setup.

The setup included:

- Wazuh server deployed using OVA (all-in-one setup)
- Core components: Wazuh Manager, Wazuh Indexer, and Wazuh Dashboard
- Integration of multiple monitored endpoints: Windows Server (Domain Controller), Windows 10 Client, and Ubuntu Client

6.2 Key Achievements

- ✓ Successfully deployed Wazuh SIEM using pre-configured OVA
- ✓ Configured network communication within the virtual lab
- ✓ Implemented port forwarding for dashboard access from host system
- ✓ Installed and configured Wazuh agents on multiple endpoints
- ✓ Verified agent connectivity and real-time log collection
- ✓ Enabled centralized monitoring through Wazuh Dashboard
- ✓ Identified and resolved practical setup challenges:
    - Network communication issues
    - Agent connectivity and configuration errors

6.3 Practical Learning Outcomes

This project provided hands-on experience with:

- SIEM architecture and components

- Centralized log collection and monitoring
- Agent-based endpoint security monitoring
- Dashboard-based security analysis
- Multi-platform monitoring (Windows and Linux)

It also provided insights into:

- How logs are collected, processed, and analyzed
- The role of detection rules in identifying suspicious activity
- Real-time visibility across multiple systems

6.4 Real-World Relevance

Wazuh and similar SIEM tools are widely used in enterprise environments for:

- Continuous security monitoring
- Threat detection and alerting
- Incident investigation and response
- Compliance and log management

This project simulates a real-world Security Operations Center (SOC) setup where multiple endpoints are monitored centrally for potential threats.

6.5 Future Work

- Simulate and detect brute-force attacks using Wazuh
- Analyze authentication logs from Active Directory
- Create custom detection rules for advanced threat detection
- Integrate additional endpoints for large-scale monitoring
- Perform incident response and alert investigation